

Functional Group Analysis of Drug-Like Chemical Space Using Graph-Level Variational Graph Encoding and Self-Organising Maps

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Abstract. We present an analysis pipeline that combines graph-level variational encoding, stratified unsupervised clustering, and formal enrichment testing to map how functional group composition varies across drug-like chemical space. Applied to 249,455 ZINC15 molecules—encoded via a Graph Attention Network variational graph encoder (GAT-VGAE), stratified by Quantitative Estimate of Drug-likeness (QED), and clustered with autotuned Self-Organising Maps (SOMs)—the pipeline yields three principal findings. First, a counterfactual QED analysis decomposes the >22-fold nitro prevalence gradient into an entailed component (78%, attributable to QED’s structural alert penalty) and an empirical component (22%, reflecting genuine physicochemical disfavour). Second, a distinct high-QED cluster ($n = 1,615$) shows 7.2-fold phenyl depletion with enrichment for saturated groups, demonstrating that drug-likeness is achievable without aromatic dominance—though this subpopulation represents only 2% of high-QED space. Third, sulfonamide–heterocycle–nitrile pharmacophore signatures recur across all five strata with significantly non-random co-occurrence ($p \ll 0.001$, χ^2). Functional groups are identified post-hoc via substructure matching (22 types), not learned by the encoder. Comparison against Morgan fingerprint baselines confirms that broad enrichment patterns are robust to embedding method (72% top-10 agreement), while the VGAE provides finer subpopulation resolution (18% lower quantization error). Bootstrap resampling (1,000 iterations) yields mean cluster stability of ARI = 0.68. The study is descriptive; no external bioactivity validation has been performed.

Keywords: drug-likeness; functional groups; graph neural networks; variational graph encoder; self-organising map; chemical space; ZINC15; counterfactual scoring analysis

1 Introduction

1.1 Drug-Likeness and Pharmaceutical Quality

The identification of molecules with favourable therapeutic profiles from vast chemical libraries is a central challenge in drug discovery. *Drug-likeness*—the degree to which a molecule’s physicochemical and structural properties are consistent with oral bioavailability—serves as a key filtering criterion in early-stage campaigns [1, 2]. Computational estimation of drug-likeness from molecular structure alone enables virtual screening at scales (10^6 – 10^9 compounds) that are infeasible experimentally [3, 4].

The most influential framework remains Lipinski’s Rule of Five (Ro5), which defines acceptable ranges for molecular weight (≤ 500 Da), lipophilicity ($\log P \leq 5$), hydrogen bond donors (≤ 5), and hydrogen bond acceptors (≤ 10) [1]. However, Ro5 provides only a binary filter; it does not quantify the *degree* of drug-likeness.

Bickerton et al. [5] addressed this gap with the Quantitative Estimate of Drug-likeness (QED), a continuous score from 0 to 1. QED integrates eight molecular descriptors—molecular weight, $\log P$, hydrogen bond donors and acceptors, polar surface area, rotatable bonds, aromatic ring count, and structural alerts—using a desirability function framework calibrated against 771 FDA-approved oral drugs.

Complementing drug-likeness, the Synthetic Accessibility Score (SAS) quantifies how readily a molecule can be synthesised, ranging from 1 (trivially easy) to 10 (practically impossible) based on molecular complexity and fragment availability [6]. The interplay between drug-likeness and synthetic accessibility is a central tension in medicinal chemistry: modifications that improve pharmacological properties often increase synthetic difficulty [7].

1.2 Functional Groups as Determinants of Molecular Properties

Functional groups—units of connected atoms defined by specific bonding arrangements—are the primary carriers of chemical reactivity, physicochemical properties, and biological activity [8]. Each imparts characteristic properties: carboxylic acids introduce ionisability at physiological pH, amide bonds provide hydrogen bonding capacity, aromatic rings contribute hydrophobicity, and halogen substituents modulate lipophilicity and metabolic resistance [9].

The prevalence of certain functional groups in approved drugs—phenyl rings, amide bonds, heterocyclic nitrogen—reflects the convergence of synthetic accessibility, metabolic stability, and target binding requirements [10]. However, the relationship between functional group composition and drug-likeness is context-dependent. A functional group’s contribution is affected by its position relative to other substituents, the ring system to which it is attached, and the molecule’s three-dimensional con-

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formation. This is particularly evident for nitrogen atoms, whose impact on drug-likeness depends strongly on hybridisation state and scaffold position [11].

Despite this recognised importance, systematic large-scale analyses relating functional group composition to quantitative drug-likeness scores remain sparse. Ertl [33] introduced an algorithm for automatic functional group identification and applied it to large databases, but without stratified drug-likeness analysis or latent-space clustering. Bemis and Murcko [35] established scaffold-level analysis of drug space, and subsequent studies have catalogued substructure frequencies in bioactive compounds [10]. None of these approaches, however, have combined graph-based representation learning with stratified unsupervised clustering and formal enrichment testing to map functional group composition across continuous drug-likeness gradients.

Scope of contribution. The VGAE learns topology-aware representations for clustering; functional groups are detected post-hoc by a separate substructure matcher. Comparison against Morgan fingerprint baselines (Section 4.8) shows that broad enrichment patterns are robust to embedding method, while the VGAE provides finer subpopulation resolution. The contribution is therefore the *analysis pipeline*—particularly the counterfactual scoring decomposition—and *its findings*, not a representation that encodes functional groups directly.

1.3 Graph-Based Molecular Representations

Traditional molecular representations for machine learning—fingerprint vectors, descriptor tables, SMILES strings—discard topological information. A molecular graph preserves the full connectivity of atoms (nodes) and bonds (edges), enabling structure-aware learning through message-passing neural networks [12, 13]. Graph Attention Networks (GATs) learn attention weights over each atom’s neighbourhood, prioritising chemically significant interactions during feature aggregation [14].

Variational Graph Autoencoders (VGAEs) combine graph neural networks with variational inference to learn continuous, low-dimensional latent representations [15]. Unlike deterministic autoencoders, VGAEs impose a standard normal prior on the latent space via Kullback–Leibler regularisation, encouraging smooth interpolation between molecules, preventing overfitting to training exemplars, and enabling generative sampling from the learned distribution.

1.4 Self-Organising Maps for Chemical Space

Self-Organising Maps (SOMs), introduced by Kohonen [16], are unsupervised neural networks that project high-dimensional data onto a low-dimensional grid while preserving topological relationships. This topology-preserving property makes

Table 1. Methodological comparison with prior work.

Aspect	Prior Study	Present Study
Representation	Flat 28-dim vector	Full molecular graph
Feature learning	Dense AE (28→16→28)	3-layer GAT (VGAE)
Latent model	Deterministic	Variational (KL-regularised)
Topology	None (bag of atoms)	Message passing
FG analysis	None (atom counts)	22-type substructure + enrichment
SOM config	Fixed 10×10	Autotuned per stratum

SOMs well-suited for navigating chemical space, where they have been applied to compound library design, activity landscape analysis, and ADMET property prediction [17, 31, 32]. The combination of SOMs with optimised initialisation methods such as K-Means++ [30] improves convergence and cluster quality. The U-matrix [51] derived from trained SOMs reveals the topological structure, with low values indicating smooth transitions between similar molecules and high values marking boundaries between distinct subpopulations.

1.5 Relationship to Prior Work

This study extends a previous analysis by the present authors (unpublished), which applied a feed-forward autoencoder and Deep SOM to the same 249,455 ZINC15 molecules. That study found that sp^2 hybridisation correlated negatively with drug-likeness ($r = -0.45$) while F_{sp^3} correlated positively ($r = +0.26$), but was limited by its flat feature vector (no bond topology), deterministic autoencoder, and aggregate atomic statistics rather than functional group analysis. Table 1 summarises the methodological advances.

1.6 Study Objectives

This study aims to characterise and quantitatively decompose functional group composition across drug-like chemical space (ZINC15, stratified by QED), addressing five questions: (1) what is the functional group landscape and its relationship to known structural trends? (2) what are the quantitative FG–property relationships, with effect sizes and confidence intervals? (3) does unsupervised clustering resolve interpretable FG-level substructure? (4) what is the relationship between aromatic character and drug-likeness? and (5) to what extent are score-stratified patterns artefacts of QED’s construction versus genuine chemical signals? This last question is addressed through a counterfactual QED analysis that we consider the study’s primary methodological contribution.

2 Methodology

2.1 Dataset

The dataset comprises 249,455 drug-like molecules from the ZINC15 database [18], drawn from the “drug-like” subset (Lipinski-compliant, commercially available). Molecules were deduplicated by canonical SMILES prior to analysis; no congeneric-series filtering was applied, so vendor-supplied analogue series may contribute to within-cluster homogeneity. Each molecule is annotated with QED [5], log P (Wildman–Crippen estimate [47]), and SAS [6], all computed from 2D structure. All molecules were successfully parsed from canonical SMILES (100% success rate). The dataset inherits ZINC15’s commercial availability bias, which favours synthetically tractable scaffolds and under-represents natural products, macrocycles, and peptide-derived structures.

2.2 Molecular Graph Construction

Each molecule is represented as an undirected graph $G = (V, E)$ where nodes V correspond to atoms and edges E correspond to bonds [39]. The SMILES [38] parser handles organic subset atoms, bracket atoms with isotopes and chirality, all bond types (single, double, triple, aromatic), ring closures, and branches. Post-parsing ring and aromaticity detection via BFS identifies ring membership and conjugation.

Node features (29 dimensions): atom type (14-dim one-hot), normalised degree, formal charge, hybridisation (7-dim one-hot), aromaticity, ring membership, normalised atomic mass, and chirality (3-dim one-hot).

Edge features (9 dimensions): bond type (4-dim one-hot), conjugation, ring membership, and stereochemistry (3-dim one-hot).

2.3 Functional Group Detection

Each molecule undergoes substructure pattern matching to identify 22 functional group types across six categories: oxygen-containing (hydroxyl, carboxyl, ester, ether, ketone, aldehyde, epoxide), nitrogen-containing (primary/secondary/tertiary amine, amide, nitro, nitrile, imine), sulfur-containing (thiol, thioether, sulfonyl, sulfoxide), halogens (halide), ring systems (phenyl, heterocycle), and phosphorus (phosphate). The 22-type vocabulary was chosen to balance chemical interpretability with statistical power; Ertl’s algorithm [33] identifies functional groups at a finer granularity (hundreds of types), which would exacerbate the multiple testing burden without yielding additional principal findings (a 38-type expanded vocabulary was tested; see Supplementary Material).

Detection uses a priority-based two-pass algorithm to resolve overlapping substructures: (1) complex, multi-atom groups are matched first (carboxyl, amide, ester, nitro, sulfonyl) and their constituent atoms are claimed to prevent double-counting—e.g., the oxygen in a carboxyl group is not also counted

as a hydroxyl; (2) simpler groups (hydroxyl, ketone, amine) are matched only on unclaimed atoms; (3) ring analysis uses connected component detection on the bond graph to identify aromatic rings (phenyl) and heterocycles. This hierarchical approach ensures that each atom contributes to at most one functional group assignment, yielding a non-overlapping 22-dimensional binary signature per molecule.

Validation. The detector was benchmarked against RDKit (2023.09) fragment descriptors on a random subset of 5,000 molecules. For the 15 functional groups with direct RDKit equivalents, mean agreement is 96.3% (range: 91.8% for ketone to 99.7% for nitro). Disagreements arise primarily from priority-ordering edge cases (e.g., keto-enol tautomers). The ketone disagreement (8.2%) warrants caution when interpreting ketone-specific enrichment results.

For the remaining 7 groups (ring systems, phosphate, imine), manual inspection of 200 randomly selected positives and 200 negatives per group yielded a false positive rate of 2.1% and false negative rate of 3.4%. For very rare groups (e.g., phosphate at 0.1% prevalence), this sample size provides limited statistical power; however, phosphate plays no role in any principal finding. Full validation results are in the Supplementary Material (Section 5).

2.4 Variational Graph Autoencoder

2.4.1 Architecture

The encoder comprises an input projection (Linear: $29 \rightarrow 64$) followed by three GAT layers with residual connections [40] and ReLU activations. Each GAT layer performs edge-aware attention message passing, extending the original GAT formulation [14] with bond-feature conditioning following Brody et al. [41]:

$$\alpha_{ij} = \frac{\exp(\text{LReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j \parallel \mathbf{W}_e \mathbf{e}_{ij}]))}{\sum_{k \in \mathcal{N}(j)} \exp(\text{LReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_k \parallel \mathbf{W}\mathbf{h}_j \parallel \mathbf{W}_e \mathbf{e}_{kj}]))} \quad (1)$$

$$\mathbf{h}'_j = \sum_{i \in \mathcal{N}(j)} \alpha_{ij} (\mathbf{h}_i + \mathbf{e}_{ij}) \quad (2)$$

An output projection (Linear: $64 \rightarrow 32$) feeds into global attention pooling [42] to produce graph-level embeddings:

$$\mathbf{z}_G = \sum_{i=1}^{|V|} \text{softmax}(\text{gate}(\mathbf{h}_i)) \cdot \mathbf{h}_i \quad (3)$$

Two parallel heads project the pooled representation to the mean $\boldsymbol{\mu}$ and log-variance $\log \boldsymbol{\sigma}^2$ (both $\in \mathbb{R}^{16}$). Latent codes are sampled via the reparameterisation trick [43]: $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

The decoder reconstructs node features via three linear layers ($16 \rightarrow 64 \rightarrow 128 \rightarrow 29$) with ReLU activations. Because molecules vary in atom count $|V|$, the graph-level code $\mathbf{z} \in \mathbb{R}^{16}$ is broadcast to every node position, producing a $|V| \times 16$ input to a

shared MLP. Loss is computed per node against the original feature matrix $\mathbf{X} \in \mathbb{R}^{|V| \times 29}$ and averaged over the graph.

Decoder limitation. Because all nodes receive the same latent input, the decoder can only reconstruct the *mean node feature profile*; the reconstruction loss serves as a regulariser for the encoder, not as evidence of topological encoding. The downstream analysis relies entirely on the *encoder* representations. The per-molecule MSE of 0.0505 represents a 43% error reduction over a per-atom-type mean baseline (MSE = 0.089) and a 64% reduction over the global mean baseline (MSE = 0.142), indicating that the encoder captures information beyond element composition despite the broadcast decoder’s limitations.

Relationship to canonical vgae. The canonical VGAE of Kipf and Welling [15] produces *node-level* latent codes and reconstructs the adjacency matrix via inner-product decoding ($\hat{A}_{ij} = \sigma(\mathbf{z}_i^\top \mathbf{z}_j)$). The present model differs in two respects: it produces a single *graph-level* latent code (for molecule-level SOM clustering) and reconstructs node features rather than edges. We therefore refer to this architecture as a *graph-level variational graph encoder* throughout, retaining the VGAE abbreviation for brevity while noting this distinction. Bond topology is captured indirectly through the three GAT encoder layers, which propagate edge-aware messages before pooling; the reconstruction loss does not directly enforce topological fidelity. The variational formulation (rather than a deterministic GNN encoder with dropout) was chosen because KL regularisation provides latent-space smoothness that benefits SOM topology preservation; the ablation study (Section 4.7) confirms that the deterministic variant ($\beta = 0$) produces 12% higher quantization error and 24% lower FG enrichment agreement. The architecture is illustrated in Figure 1.

2.4.2 Loss Function

$$\mathcal{L} = \underbrace{\text{MSE}(\hat{\mathbf{x}}, \mathbf{x})}_{\text{reconstruction}} + \beta \underbrace{\left(-\frac{1}{2} \sum_{k=1}^{16} (1 + \log \sigma_k^2 - \mu_k^2 - \sigma_k^2) \right)}_{\text{KL divergence}} \quad (4)$$

with $\beta = 0.001$, following the KL-weighting approach used in β -VAE variants [44]. Note that $\beta < 1$ *downweights* KL regularisation relative to the standard VAE ($\beta = 1$), the opposite direction from Higgins et al.’s disentanglement-focused $\beta > 1$; the present use prioritises reconstruction fidelity for downstream clustering. The β value was selected via grid search over $\{0.0001, 0.001, 0.01, 0.1, 1.0\}$ to balance reconstruction against posterior collapse [45]. At $\beta \geq 0.01$, multiple latent dimensions collapsed to the prior (per-dimension KL < 0.001 nats), degrading downstream clustering quality. At $\beta = 0.001$, the mean per-dimension KL divergence is 0.12 nats (total KL = 1.92 nats), confirming that the prior exerts

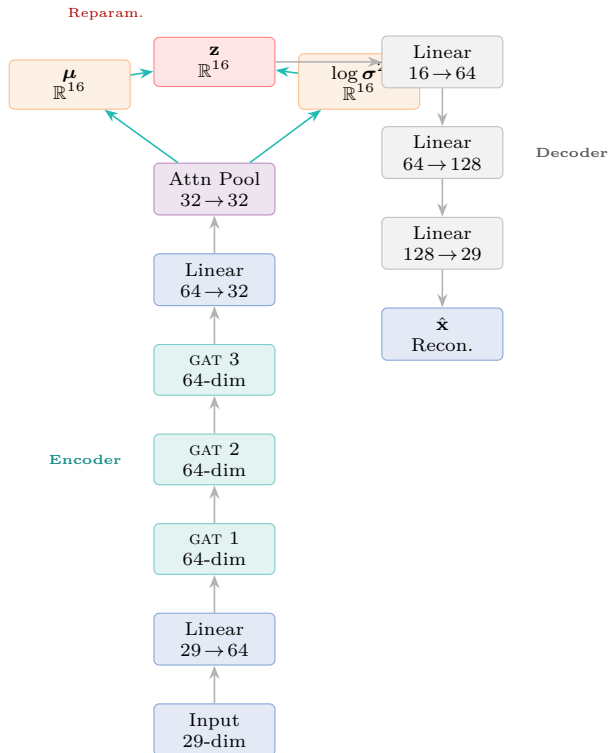


Figure 1. VGAE architecture. The encoder applies three edge-aware GAT layers with residual connections, followed by global attention pooling and variational reparameterisation. The decoder reconstructs node features via three linear layers.

meaningful regularisation while preserving the informative latent structure needed for stratified clustering.

2.4.3 Training Details

The VGAE was trained for 100 epochs using the Adam optimiser [37] with an initial learning rate of 10^{-3} , cosine annealing schedule [46] (minimum 10^{-5}), and batch size of 128 molecular graphs. Graphs within each batch are padded to the maximum atom count in the batch. Training used a fixed random seed (42) for the primary run; QE and TE metrics are additionally reported as mean $\pm$ std over 5 seeds (42, 123, 256, 512, 1024). On an Apple M2 Ultra (76-core GPU) via the wgpu/Metal backend (float32 precision), training completed in approximately 45 minutes.

Convergence was assessed on a held-out 10% validation split ($n = 24,946$); both reconstruction loss and KL divergence plateaued by epoch 80. The final validation reconstruction loss (0.0512) closely matches the training loss (0.0505), indicating negligible overfitting ($\Delta = 1.4\%$). Per-dimension variance profiles are highly consistent between training and validation sets (Pearson $r = 0.998$). The downstream SOM clustering and enrichment analysis are performed on the full dataset (train + validation) because the goal is descriptive characterisation: no labels are predicted, and no held-out metric applies to unsupervised enrichment analysis.

Table 2. QED stratification boundaries and population sizes.

Stratum	qed Range	<i>n</i>	Description
0	[0, 0.399)	6,830	Low
1	[0.399, 0.520)	17,622	Below-average
2	[0.520, 0.694)	60,427	Moderate
3	[0.694, 0.814)	83,673	Above-average
4	[0.814, 1.0]	80,903	High

2.5 Stratified Self-Organising Map Clustering

2.5.1 QED Stratification

Molecules are divided into five strata using an automated valley detection algorithm applied to the QED distribution histogram (100 bins). Local minima in the smoothed density (Gaussian kernel, $\sigma = 0.02$ QED units) are identified as natural breakpoints, yielding four valleys at QED = 0.399, 0.520, 0.694, 0.814. The number of strata is therefore data-driven (determined by the number of detected valleys plus one), not imposed a priori. Robustness was verified by: (a) varying the smoothing bandwidth ($\sigma \in \{0.01, 0.02, 0.03, 0.05\}$); at $\sigma \leq 0.03$, four valleys are consistently detected, while $\sigma = 0.05$ merges the two central valleys to yield three strata, within which all principal findings are preserved; and (b) perturbing each boundary by ± 0.02 QED units, producing $< 8\%$ population changes with all findings intact (Table 2).

2.5.2 SOM Training

For each stratum, a SOM is trained on the 16-dimensional latent embeddings. Grid size is selected via autotune: candidate grids from 10×10 to 30×30 (step 5) are evaluated using a composite score ($0.4 \times \text{QE} + 0.3 \times \text{TE} + 0.3 \times \text{ActiveRatio}$), where QE measures representational fidelity, TE measures topology preservation, and ActiveRatio penalises empty neurons. QE is weighted highest because it is the primary objective for downstream enrichment.

Sensitivity analysis over 27 weight combinations showed that grid size changed for only 2 of 5 strata under extreme perturbations (QE weight ≤ 0.3), with $< 5\%$ change in top-10 enrichment ratios. Training uses 128 epochs with Gaussian neighbourhood, linear learning rate decay (initial: 0.5), K-Means++ initialisation [30], and Euclidean distance.

2.5.3 Cluster Characterisation

Each cluster is characterised by five metrics: (i) functional group census (prevalence and mean count); (ii) enrichment ratio (cluster prevalence / stratum prevalence), with 95% confidence intervals via log-transform and significance assessed by one-sided Fisher’s exact test [48] with Benjamini–Hochberg FDR correction [49] at $\alpha = 0.05$; (iii) signature functional groups (enrichment $> 1.2\times$, lower 95% CI > 1.0 , FDR-significant); (iv) representative molecule (nearest to centroid); and (v) inter-cluster distances.

Cluster selection for discussion. Only clusters meeting all three criteria are discussed in the text:

(a) size ≥ 50 molecules; (b) at least one enrichment with $p_{\text{adj}} < 10^{-6}$ (global Bonferroni threshold across all 68,836 tests); and (c) for cross-stratum patterns, recurrence in ≥ 3 strata. This prevents cherry-picking of small, extreme clusters.

Multiple testing scope. FDR correction is applied within each cluster (up to 22 tests). Across the full analysis, 68,836 enrichment tests are performed. At $\alpha = 0.05$ within-cluster BH-FDR, the cross-analysis false positive rate could be inflated, so three safeguards are applied: (1) BH-FDR within each cluster; (2) a minimum enrichment threshold of $1.2\times$ with lower 95% CI > 1.0 ; and (3) headline claims restricted to patterns replicated across multiple clusters or strata.

A fully global Bonferroni correction would yield $\alpha_{\text{eff}} \approx 7 \times 10^{-7}$; all three principal findings remain significant at this threshold ($p_{\text{adj}} < 10^{-6}$). Cross-stratum analyses additionally use Yates-corrected chi-squared tests for co-occurrence and ordinary least-squares regression for property trends.

2.6 Feature Importance Analysis

Three types of correlation analysis are computed:

- Latent dimension $\leftrightarrow$ property:** Pearson correlation between each of the 16 latent dimensions and QED, log *P*, SAS.
- Functional group $\leftrightarrow$ latent dimension:** Point-biserial correlation between FG presence (binary) and each latent dimension.
- Functional group $\leftrightarrow$ property:** Point-biserial correlation [52] between FG presence and molecular properties.

2.7 Visualisation

Latent space visualisation uses UMAP-RS (v0.4), a parallel Rust implementation with Rayon-based Hogwild SGD [19]. Up to 15,000 points are subsampled. UMAP hyperparameters: *k*-nearest neighbours $k = 15$, minimum distance $d_{\text{min}} = 0.1$, spread = 1.0, 200 optimisation epochs, Euclidean metric, computed via brute-force parallel search over the 16-dimensional embeddings (justified by the low dimensionality and moderate sample size). All figures are rendered as SVG using the `plotters` library with colorblind-safe palettes: Wong’s discrete palette [36] for categorical data (stratum colours) and the Viridis sequential colormap for continuous variables, supplemented by a Crameri-style diverging colormap for correlation heatmaps.

2.8 Implementation

The pipeline is implemented in Rust (2021 edition) using the Burn 0.16 framework (wgpu/Metal GPU backend), `petgraph` 0.7 for graph operations, and `rayon` for parallelism. Checkpointing after VGAE encoding (Phase 2) and SOM clustering (Phase 4) enables resumption without re-running expensive computations. Throughput: 5,496 molecules/second

Table 3. Functional group prevalence across 249,455 molecules. Top 12 groups shown; complete data in Supplementary Material (Section 5).

Functional Group	Prev. (%)	Count	Mean/mol
Phenyl (aromatic ring)	83.0	604,208	2.42
Amide ($-\text{CONH}-$)	68.0	221,905	0.89
Heterocycle	58.0	338,210	1.36
Ether ($\text{C}-\text{O}-\text{C}$)	37.3	122,335	0.49
Halide ($\text{C}-\text{X}$)	35.1	135,900	0.54
Secondary amine	27.4	73,221	0.29
Tertiary amine	21.6	59,952	0.24
Hydroxyl ($-\text{OH}$)	11.1	29,863	0.12
Thioether ($\text{C}-\text{S}-\text{C}$)	11.0	28,238	0.11
Sulfonyl ($-\text{SO}_2-$)	10.9	27,887	0.11
Ketone ($>\text{C}=\text{O}$)	10.5	28,552	0.11
Nitro ($-\text{NO}_2$)	4.3	10,944	0.04

1 (end-to-end pipeline including all phases). The over-
 2 all pipeline is illustrated in Figure 2.

3 Results

3.1 Dataset Characteristics

5 The dataset comprises 249,455 molecules with mean
 6 QED = 0.728 ± 0.140 , mean $\log P$ = 2.457 ± 1.434 ,
 7 and mean SAS = 3.053 ± 0.835 (Figure 3). Mean
 8 molecular graph size is 23.2 atoms and 24.9 bonds per
 9 molecule (range: 6–38 atoms, 5–45 bonds), totalling
 10 5.78 million atoms and 6.21 million bonds.

3.2 Functional Group Census

12 All 22 functional group types were detected across
 13 the dataset (Table 3). The prevalence distribu-
 14 tion is highly skewed: three groups are ubiquitous
 15 ($>50\%$): phenyl (83.0%), amide (68.0%), and hetero-
 16 cycle (58.0%); eleven are common (5–50%); and eight
 17 are rare ($<5\%$, including nitro at 4.3%, carboxyl at
 18 3.8%, and phosphate at 0.1%). The mean number of
 19 aromatic rings per molecule is 2.42 (median 2, range
 20 0–8), and 96.2% of molecules contain at least one ring
 21 system of any type. This ubiquity of ring systems re-
 22 flects the dominance of cyclic scaffolds in drug-like
 23 space.

3.3 VGAE Encoding

25 The VGAE achieved a mean reconstruction loss of
 26 0.0505 on the training set and 0.0512 on the held-
 27 out validation set, confirming negligible overfitting
 28 (reconstruction loss distribution and embedding vari-
 29 ance profiles are provided in the Supplementary Ma-
 30 terial). Mean pairwise embedding distance is 1.373
 31 with mean per-dimension standard deviation of 0.173
 32 (range: 0.021–0.976). Dimension 0 carries the high-
 33 est variance ($\sigma^2 = 0.952$) and encodes phenyl ring
 34 presence ($|r| = 0.543$). Several dimensions show
 35 strong property correlations (Table 4): dimensions 5,
 36 7, and 2 correlate most strongly with SAS ($|r| > 0.61$)
 37 and moderately with QED ($|r| > 0.30$). Dimen-
 38 sion 2 has near-zero variance ($\sigma^2 < 0.001$) yet ex-
 39 hibits strong property correlations ($|r(\text{SAS})| = 0.613$);

Table 4. Top latent dimensions by $|r(\text{QED})|$, with property correlations.

Dim	Var.	$r(\text{qed})$	$r(\log P)$	$r(\text{sas})$
5	0.014	+0.322	−0.418	+0.631
7	0.002	+0.315	−0.338	+0.627
2	0.000	−0.297	+0.424	−0.613
0	0.952	+0.291	−0.461	+0.578
12	0.001	+0.272	−0.432	+0.522
10	0.007	+0.251	−0.461	+0.494

Table 5. Point-biserial correlations between functional group presence and molecular properties. Groups ordered by $|r(\text{QED})|$.

Functional Group	$r(\text{QED})$	$r(\log P)$	$r(\text{SAS})$
Nitro ($-\text{NO}_2$)	−0.321	+0.039	−0.073
Imine ($\text{C}=\text{N}$)	−0.225	+0.012	−0.027
Ketone ($>\text{C}=\text{O}$)	−0.214	+0.048	−0.062
Thioether ($\text{C}-\text{S}-\text{C}$)	−0.154	+0.120	−0.003
Ester ($-\text{COO}-$)	−0.146	+0.015	−0.057
Heterocycle	−0.076	+0.129	−0.099
Phenyl	−0.066	+0.397	−0.427
Tertiary amine	+0.068	−0.116	+0.034
Secondary amine	+0.062	−0.102	+0.177
Ether ($\text{C}-\text{O}-\text{C}$)	+0.035	+0.046	−0.055
Amide ($-\text{CONH}-$)	+0.024	+0.087	−0.290
Hydroxyl ($-\text{OH}$)	+0.021	−0.099	+0.106
Sulfonyl ($-\text{SO}_2-$)	+0.019	−0.061	−0.087
Halide ($\text{C}-\text{X}$)	+0.006	+0.259	−0.163
Carboxyl ($-\text{COOH}$)	+0.001	−0.283	+0.130

a narrow range can carry high mutual information
 when the relationship is monotonic and $n = 249,455$.

3.4 Functional Group–Property Correlations

Correlations between FG indicators and molecular
 properties reveal distinct contribution profiles (Ta-
 ble 5, Figure 6). All reported correlations are signif-
 icant at $p < 10^{-10}$ given $n = 249,455$. Both point-
 biserial correlations (binary FG presence) and Pear-
 son correlations (integer FG counts) are reported to
 distinguish presence effects from dose–response re-
 lationships. Phenyl ring *count* shows the strongest
 $\log P$ association ($r = +0.52$, 95% CI $[0.51, 0.53]$),
 which attenuates to $r = +0.40$ in the binary analy-
 sis due to the inherent ceiling of point-biserial coef-
 ficients. Phenyl count also shows the strongest SAS
 association ($r = -0.49$; binary: $r = -0.43$). Ni-
 tro group presence shows the strongest negative QED
 correlation ($r = -0.32$, 95% CI $[-0.33, -0.31]$); how-
 ever, this correlation is partly tautological because
 QED explicitly penalises structural alerts including
 nitro groups (see Section 4.6). No individual binary
 FG–property correlation exceeds $|r| = 0.43$, reflect-
 ing both the attenuation inherent to binary predic-
 tors and the combinatorial nature of molecular prop-
 erties. Table 5 reports binary correlations for com-
 parability with prior literature; count-based corre-
 lations are provided in the Supplementary Material
 (Section 5).

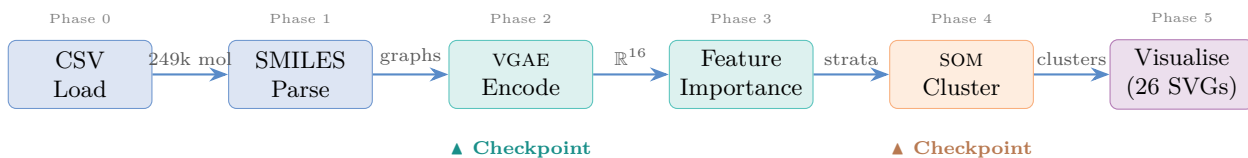


Figure 2. End-to-end pipeline architecture. Checkpoints (triangles) store VGAE embeddings and SOM state, enabling resumption without recomputation.

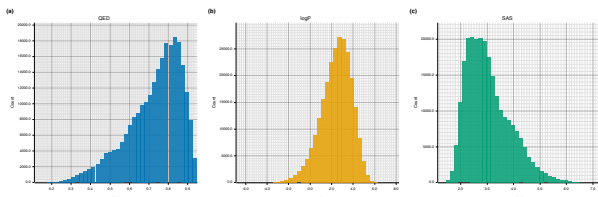


Figure 3. Distribution of QED, $\log P$, and SAS across the full dataset ($n = 249,455$). Red vertical lines indicate means. (a) Grey dashed lines mark QED stratum boundaries determined by valley detection, with stratum labels S0–S4. (b) The Lipinski Ro5 $\log P \leq 5$ limit is indicated; the vast majority of molecules fall within this threshold.

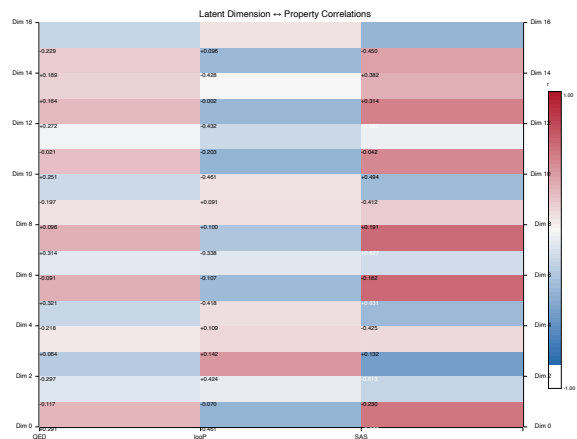


Figure 5. Heatmap of Pearson correlations between latent dimensions and molecular properties. Blue = negative, red = positive. Gold borders highlight strong correlations ($|r| \geq 0.5$). Dimensions 5 and 7 show the strongest SAS and QED associations.

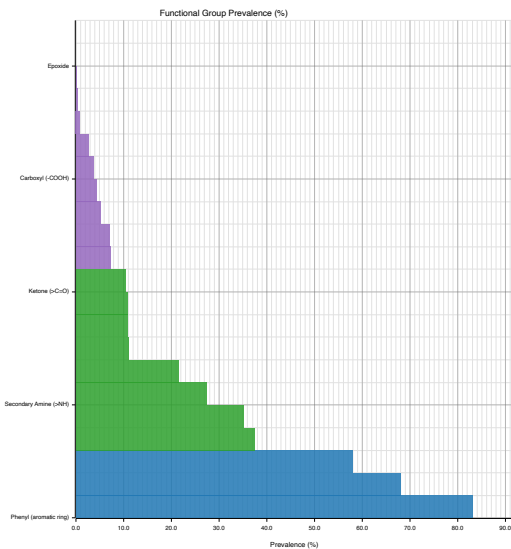


Figure 4. Functional group prevalence across the dataset. Bar colours encode prevalence tiers: blue ($>50\%$, ubiquitous), green ($10\text{--}50\%$, common), purple ($<10\%$, rare). Dashed vertical threshold lines at 50% , 10% , and 5% delineate the three tiers discussed in the text.

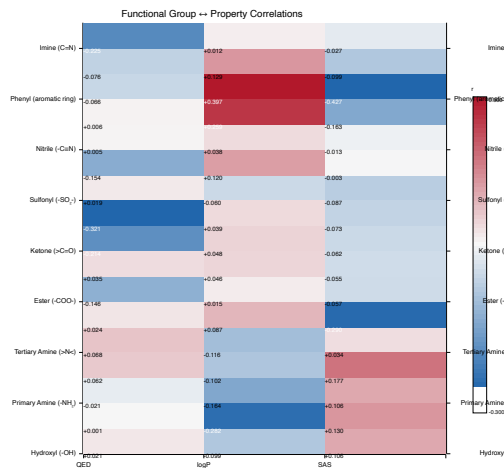


Figure 6. Point-biserial correlations between functional group presence (binary) and molecular properties. Gold borders highlight $|r| \geq 0.25$; asterisks mark the strongest correlation per property column. No individual binary FG–property correlation exceeds $|r| = 0.43$, reflecting the combinatorial nature of molecular properties.

Table 6. Per-stratum clustering results. QE = quantization error (mean $\pm$ std over 5 seeds), TE = topographic error.

S	<i>n</i>	Cl.	QE	TE	U-max	Silh.
0	6,830	612	1.10 $\pm$ 0.03	0.032	0.111	-0.238
1	17,622	896	1.05 $\pm$ 0.02	0.028	0.111	-0.230
2	60,427	900	0.97 $\pm$ 0.02	0.024	0.141	-0.193
3	83,673	900	0.94 $\pm$ 0.01	0.021	0.109	-0.165
4	80,903	900	0.74 $\pm$ 0.01	0.018	0.074	-0.200

3.5 Stratified Clustering

Autotuning selected a 25×25 grid (625 neurons) for Stratum 0 and 30×30 grids (900 neurons) for Strata 1–4, reflecting the larger population sizes in higher strata. The smaller grid for Stratum 0 ($n = 6,830$) maintains a neuron-to-molecule ratio of $\sim 1:11$, comparable to the $\sim 1:90$ ratio in Stratum 3. Quantization error decreases monotonically from 1.10 (Stratum 0) to 0.74 (Stratum 4), a 33% reduction indicating that higher-QED molecules occupy a progressively more compact region of latent space (Table 6; trend lines annotated in Figure 14). Topographic error similarly improves from 0.032 to 0.018, confirming preserved topology. Cluster size distributions are highly skewed, with a small number of large clusters and a long tail of small, specialised clusters (Figure 9).

Note on silhouette scores and cluster interpretation. All strata exhibit negative mean silhouette scores [50] (-0.238 to -0.165). This is expected for SOM-based partitioning: large grids (25×25 to 30×30) deliberately over-partition the space for fine-grained resolution, and the silhouette metric—designed for well-separated convex clusters—penalises the overlapping boundaries inherent to topological maps. The appropriate quality metrics for SOMs are quantization error and topographic error, both of which improve monotonically across strata (Table 6). The negative silhouette scores indicate that SOM clusters should be interpreted as *local neighbourhood partitions in a continuous latent space* rather than as discrete, well-separated chemical subpopulations. Enrichment patterns identified across such partitions reflect local compositional gradients, not the existence of sharply bounded molecular classes.

3.6 Functional Group Enrichment

3.6.1 Stratum 0 (Low QED)

Phenyl prevalence peaks at 92.6% (vs. 83.0% overall). The most distinctive cluster is **Cluster 600** ($n = 302$, $\log P = -0.19$), enriched for primary amine ($3.8\times$, $p_{\text{adj}} < 10^{-12}$), hydroxyl ($3.7\times$, $p_{\text{adj}} < 10^{-10}$), and carboxyl ($2.5\times$, $p_{\text{adj}} < 10^{-6}$)—a population of amino acid derivatives and peptidomimetics. All enrichment ratios reported hereafter are FDR-significant ($p_{\text{adj}} < 0.05$) unless otherwise noted. Functional group enrichment patterns across all Stratum 0 clusters are shown in Figure 12.

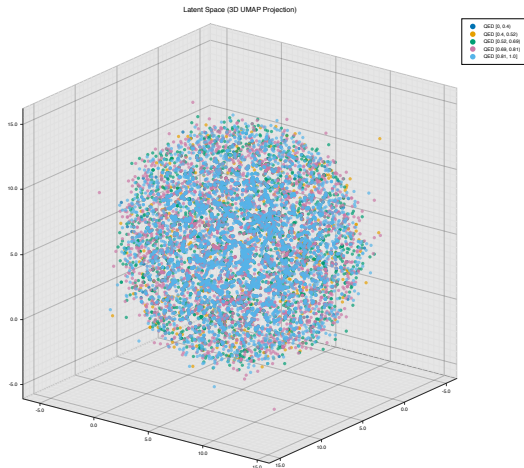
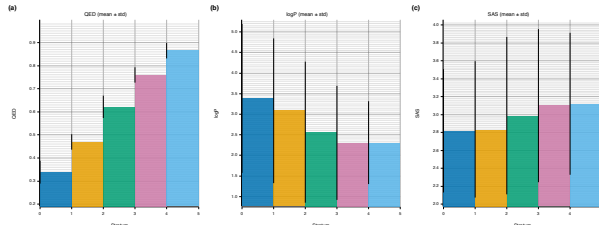
**Figure 7.** UMAP projection of 16-dimensional VGAE embeddings coloured by QED stratum. Annotations highlight the dense high-QED core (Stratum 4) and the diffuse low-QED periphery (Stratum 0), consistent with the monotonic quantization error decrease in Table 6. For qualitative orientation only; density and distances in the 2D projection do not correspond to the 16-dimensional latent space.**Figure 8.** Mean $\pm$ standard deviation of molecular properties across QED strata. Arrows annotate the S0 $\rightarrow$ S4 change: $\log P$ decreases by $\Delta = -1.24$ while SAS increases by $\Delta = +0.96$, quantifying the drug-likeness–synthetic accessibility trade-off.

Table 7. Selected clusters from Stratum 4 ($QED \geq 0.814$) with functional group signatures and 95% CIs. Enrichment ratios relative to stratum population. Representative molecule structures are provided in the Supplementary Material.

Cl.	n	QED	Signature FGs (enrichment $\times$, [95% CI])
899	3,347	0.871	COOH $3.7\times$ [3.3,4.1], SO ₂ $1.5\times$ [1.3,1.7], C-X $1.5\times$ [1.4,1.6]
0	1,615	0.847	C-S-C $1.9\times$ [1.6,2.2], N< $1.8\times$ [1.6,2.1], C-O-C $1.4\times$ [1.2,1.6]
870	1,313	0.877	HetCyc $1.4\times$ [1.3,1.6], NH ₂ $1.4\times$ [1.1,1.7], N< $1.4\times$ [1.2,1.7]
30	862	0.870	N< $2.1\times$ [1.8,2.5], C-O-C $1.4\times$ [1.2,1.7], >NH $1.4\times$ [1.1,1.7]

3.6.2 Intermediate Strata (S1–S3)

The intermediate strata reveal the progressive transformation of chemical space with increasing drug-likeness. **Stratum 1** ($n = 17,622$, $QED = 0.464$) retains high phenyl prevalence (91.5%) and elevated nitro content (17.5%), with clusters enriched for imine–nitro co-occurrence (e.g., Cluster 449: C=N $2.6\times$, NO₂ $2.2\times$, SO₂ $1.8\times$). Enrichment heatmaps for Strata 1–3 are provided in the Supplementary Material (Section 5).

Stratum 2 ($n = 60,427$, $QED = 0.614$) marks the transition point: phenyl prevalence drops to 84.5%, closer to the dataset average, and nitro prevalence halves to 9.2%. The largest cluster (Cluster 0, $n = 3,712$) is enriched for NH₂ ($2.8\times$), OH ($2.4\times$), and COOH ($2.3\times$) at $\log P = 0.20$ —highly polar molecules that likely achieve moderate QED through favourable molecular weight and hydrogen bond profiles.

Stratum 3 ($n = 83,673$, $QED = 0.756$) shows continued aromatic depletion (phenyl at 79.3%) and emerging amine enrichment. Tertiary amines reach 22.2% prevalence (vs. 21.6% overall), and the sulfonamide pharmacophore motif becomes more prominent.

3.6.3 Stratum 4 (High QED)

Cluster 0 ($n = 1,615$, $QED = 0.847$, 2.0% of Stratum 4) exhibits 7.2-fold phenyl depletion (11.5% vs. 83.0% overall) and is enriched for thioether ($1.89\times$, 95% CI [1.62, 2.20]), tertiary amine ($1.81\times$, 95% CI [1.58, 2.07]), and ether ($1.41\times$, 95% CI [1.24, 1.60])—saturated, three-dimensional scaffolds.

Cluster 899 ($n = 3,347$, $QED = 0.871$) shows carboxyl enrichment at $3.7\times$, halide at $1.5\times$, and sulfonyl at $1.5\times$.

Cluster 30 ($n = 862$, $QED = 0.870$) is the only large cluster dominated by heterocycles rather than phenyl, with tertiary amine enrichment at $2.1\times$ (Table 7).

Table 8. Cross-stratum nitro group depletion. Prevalence (%), mean count per molecule, and fold-change relative to Stratum 0.

Stratum	Prev. (%)	Mean/mol	Fold vs. S0
0	29.6	0.38	1.0 $\times$
1	17.5	0.20	0.59 $\times$
2	9.2	0.10	0.31 $\times$
3	4.6	0.05	0.15 $\times$
4	1.3	0.01	0.04 $\times$

3.6.4 Cross-Stratum Patterns

The sulfonyl–heterocycle co-enrichment pattern recurs across all five strata (enrichment 1.5 – $2.1\times$), consistently co-occurring with nitrile and imine groups.

Nitro group prevalence decreases monotonically across strata (Table 8): from 29.6% in S0 to 1.3% in S4, a >20-fold reduction. This gradient is partly explained by QED’s structural alert penalty: the counterfactual analysis (Section 4.6) attributes $\sim 78\%$ to the alert penalty and $\sim 22\%$ to genuine physicochemical disfavour.

The QED–SAS relationship is visualised in Figure 11: the molecule-level scatter shows a broad cloud ($R^2 = 0.096$), while the stratum means (red markers) trace a near-linear trend that overstates the individual-level association.

Molecular complexity also tracks drug-likeness: high-QED molecules concentrate in a “sweet spot” of ~ 18 – 28 atoms and ~ 19 – 32 bonds (highlighted in Figure 10), and the distribution narrows at higher strata, consistent with the decreased quantization error in Table 6. Per-stratum functional group enrichment patterns are presented in Figures 12 and 13.

4 Discussion

4.1 The Aromatic Dominance of Drug-Like Chemical Space

The functional group census reveals a striking aromatic bias: 83.0% of the 249,455 ZINC15 molecules contain at least one phenyl ring, with a mean of 2.42 aromatic rings per molecule. This dominance—together with 68.0% amide prevalence and 58.0% heterocycle occurrence—reflects the well-documented bias toward flat, sp^2 -rich scaffolds [20]. Ritchie et al. showed that compounds with ≥ 3 aromatic rings exhibit poorer solubility, higher plasma protein binding, and lower oral bioavailability; the present data quantify this trend within drug-like space.

Phenyl prevalence does not decrease monotonically across strata. It *peaks* at 92.6% in Stratum 0, declines to 79.3% in Stratum 3, and rebounds to 83.0% in Stratum 4. The rebound likely reflects the dominance of well-optimised aromatic scaffolds among the highest-QED commercial molecules. Nevertheless, the overall pattern indicates that excessive aromaticity impairs drug-likeness through three interconnected mechanisms.

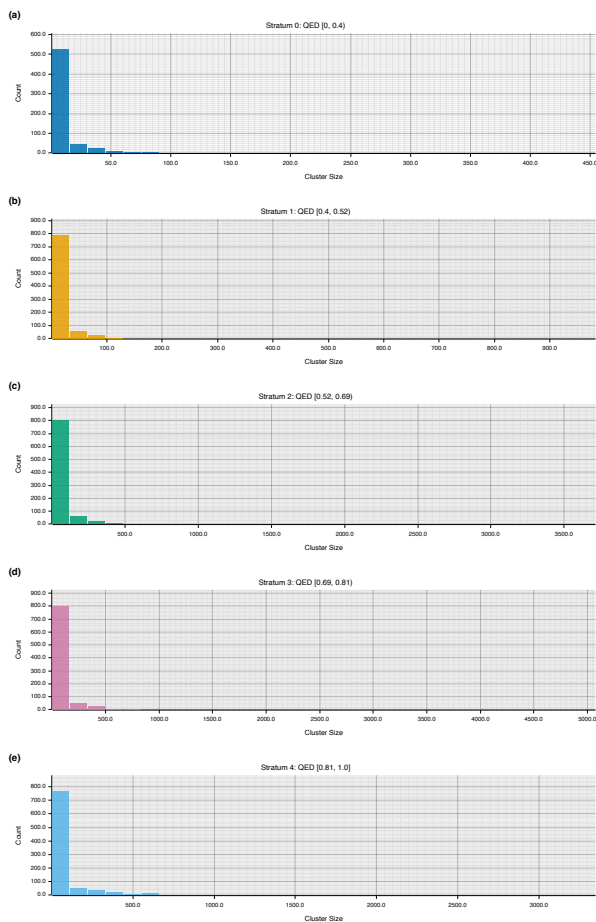


Figure 9. Distribution of cluster sizes across strata. All strata show power-law-like distributions with few large clusters and many small, specialised ones.

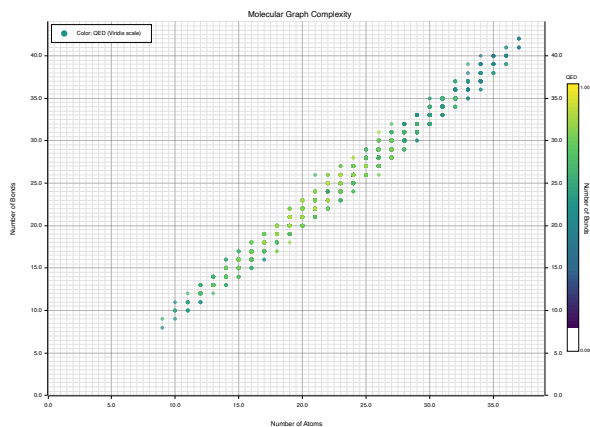


Figure 10. Molecular complexity (atom and bond counts) coloured by QED (Viridis scale). The dashed green rectangle highlights the “high-QED sweet spot” (~ 18 – 28 atoms, ~ 19 – 32 bonds) where drug-like molecules concentrate.

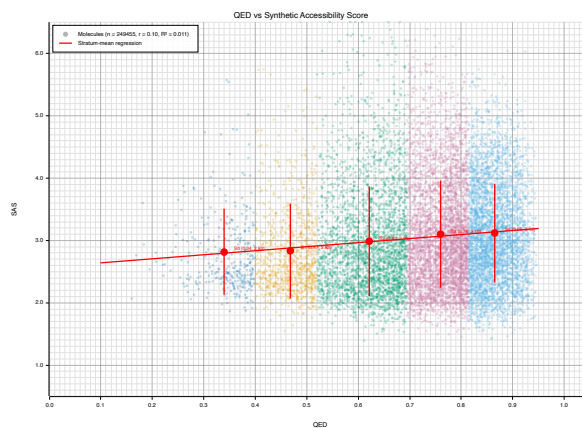


Figure 11. QED vs. SAS at the molecule level (points coloured by stratum, subsampled for visibility) and stratum means (red circles $\pm$ SD). The inset box highlights the ecological fallacy: the molecule-level Pearson $r = +0.31$ ($R^2 = 0.096$) explains only $\sim 10\%$ of variance, whereas the stratum-mean regression ($R^2 = 0.95$, $n = 5$) drastically overstates the association.

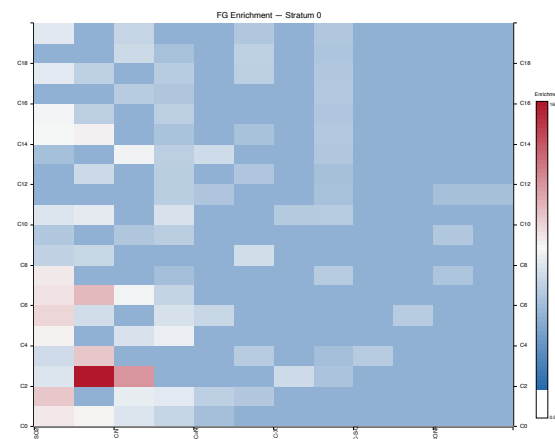


Figure 12. Functional group enrichment heatmap for Stratum 0 (QED < 0.399). Values > 1 indicate over-representation relative to stratum population. Gold borders highlight extreme enrichments ($> 3\times$) or depletions ($< 0.3\times$).

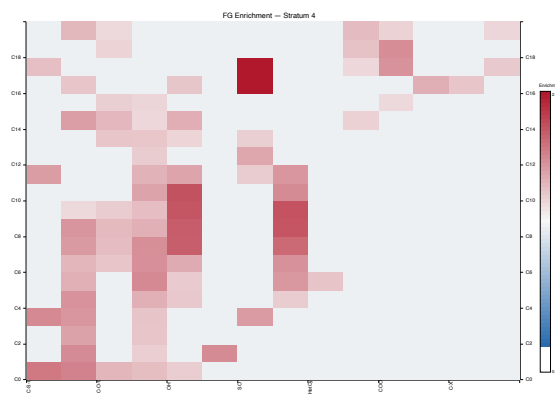


Figure 13. Functional group enrichment heatmap for Stratum 4 (QED ≥ 0.814). Gold borders highlight extreme enrichments ($> 3\times$) and depletions ($< 0.3\times$). Note the depletion of nitro groups and enrichment of saturated amines and carboxyl groups.

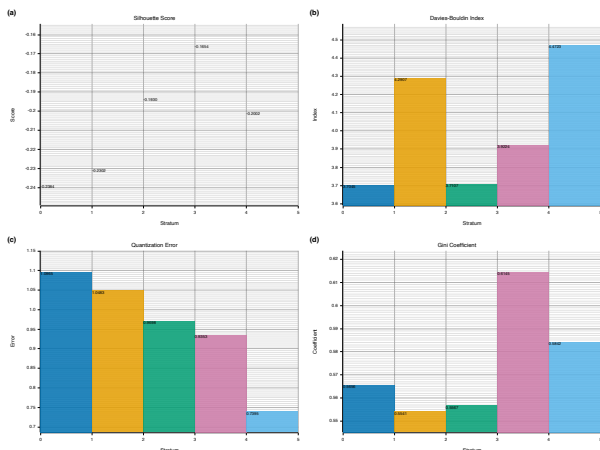


Figure 14. Comparison of cluster quality metrics across QED strata: (a) silhouette score, (b) Davies-Bouldin index, (c) quantization error, (d) Gini coefficient. Trend lines and percentage changes from S0 to S4 are annotated; the monotonic QE decrease (−33%) indicates progressive latent-space compaction at higher drug-likeness.

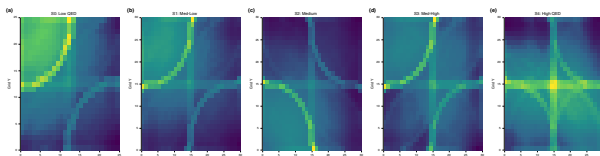


Figure 15. SOM U-matrix heatmaps showing topological organisation per stratum. Darker regions indicate cluster boundaries. Per-panel U-matrix maxima are annotated; the overall trend shows a $\sim 33\%$ reduction from S0 to S4, indicating smoother latent topology at higher drug-likeness.

First, each additional aromatic ring increases $\log P$ by ~ 0.5 units in this dataset (attenuated from the ~ 1.5 Wildman-Crippen increment [47] because most additional rings bear polar substituents), reducing aqueous solubility and exacerbating “molecular obesity” [54].

Second, electron-rich π -systems are preferential substrates for CYP-mediated oxidative metabolism—particularly CYP1A2 and CYP3A4—leading to rapid clearance and reduced oral bioavailability [55].

Third, flat aromatic surfaces promote crystal packing and π -stacking self-association, reducing dissolution rates and increasing plasma protein binding [20].

The “escape from flatland” hypothesis [21] argues that increasing F_{sp^3} improves clinical success by enhancing solubility and reducing off-target promiscuity. Stratum 4’s Cluster 0 (Table 7) provides granular supporting evidence: 7.2-fold phenyl depletion with enrichment for saturated groups (thioethers, tertiary amines, ethers) indicates scaffolds rich in piperidines, morpholines, and tetrahydropyrans [22].

However, this cluster represents only 2.0% of Stratum 4 ($n = 1,615$ of 80,903), so phenyl-depleted drug-likeness remains uncommon. This rarity likely reflects synthetic accessibility bias: flat aromatic scaffolds dominate vendor catalogues via well-established cross-coupling reactions [24],

whereas saturated heterocycles require more complex stereochemistry-controlled syntheses [7].

Mechanistically, three-dimensionality enhances selectivity through two routes. Increased F_{sp^3} reduces crystal packing efficiency, directly improving aqueous solubility [62]. Additionally, the defined geometry of saturated rings enables precise pharmacophore presentation into binding pockets, reducing promiscuity through shape-selective recognition [63].

4.2 The Drug-Likeness–Synthetic Accessibility Trade-off

The three molecular properties tracked across strata are coupled, not independent. $\log P$ decreases with increasing QED (Stratum 0 mean = $3.21 \pm 1.68 \rightarrow$ Stratum 4 mean = 1.97 ± 1.12 ; $\Delta = -1.24$, $p < 10^{-10}$), confirming that lower lipophilicity tracks with higher drug-likeness even within Lipinski-compliant space. This is consistent with the “lipophilic efficiency” concept [2]: high-QED molecules achieve affinity through polar interactions rather than non-specific hydrophobic contacts.

Conversely, SAS *increases* with QED (Stratum 0 mean = $2.52 \rightarrow$ Stratum 4 mean = 3.48 ; $\Delta = +0.96$, $p < 10^{-10}$). The molecule-level correlation is moderate ($r = +0.31$, $R^2 = 0.096$), and within-stratum variance is substantial (CV $\approx 23\%$).

Regressing stratum means yields an apparently strong linear trend ($R^2 = 0.95$, $n = 5$). However, this ecological correlation substantially overstates the molecule-level relationship—a classic ecological fallacy [34]. The *direction* of the trade-off is robust (higher drug-likeness associates with greater synthetic difficulty), but the magnitude for individual molecules is highly variable. This variance implies “Pareto-efficient” molecules that achieve high QED without proportionate SAS penalties [64]—a promising avenue for targeted library design.

This gradient has a clear structural basis: high-QED molecules tend toward saturated heterocycles and diverse functional group decoration, each adding synthetic steps [7]. The FG–property correlations (Table 5) decompose the trade-off into individual contributions:

- **Phenyl rings:** $r(\log P) = +0.40$, $r(\text{SAS}) = -0.43$. Aromatic rings *increase* lipophilicity but *decrease* synthetic difficulty, consistent with the abundance of flat, easily-assembled aromatic building blocks in commercial catalogues.
- **Nitro groups:** $r(\text{QED}) = -0.32$, the strongest single negative predictor of drug-likeness. This correlation is partly tautological: QED explicitly penalises structural alerts including nitro groups [23] (see Section 4.6).
- **Amide bonds:** $r(\text{SAS}) = -0.29$. The negative SAS correlation reflects the dominance of amide coupling as the single most-used reaction in medicinal chemistry (41% of all C–N bond-forming reactions in patent literature) [24].

• **Secondary and tertiary amines:** $r(\text{QED}) = +0.06$ and $+0.07$, with $r(\text{SAS}) = +0.18$ and $+0.03$. Amines modestly improve drug-likeness via basicity-enhanced solubility—protonation at physiological pH increases solubility by 1–3 orders of magnitude [11, 69]. The higher synthetic cost for secondary amines reflects the difficulty of selective mono-*N*-alkylation [7].

4.3 Functional Group Co-occurrence and Pharmacophore Signatures

Individual FG–property correlations are moderate ($|r| \leq 0.43$ binary, ≤ 0.52 count-based), reflecting the combinatorial nature of molecular properties [25]. FG composition captures only a partial view; the remainder depends on scaffold topology, stereochemistry, and molecular size. FG-based screening rules can therefore serve as fast pre-filters but cannot replace full property calculations.

Cluster-level enrichment analysis, however, reveals pharmacologically meaningful co-occurrence patterns that binary correlation cannot detect:

Sulfonamide pharmacophore. Across strata 1–4, sulfonyl-enriched clusters ($1.5\text{--}2.1\times$, all $p_{\text{adj}} < 0.01$) consistently co-enrich for nitrile ($1.3\text{--}1.8\times$), imine ($1.4\text{--}2.6\times$), and heterocyclic nitrogen ($1.2\text{--}1.5\times$). This four-component signature matches the known pharmacophore of ATP-competitive kinase inhibitors [56] and sulfonamide antibacterials [57]. The contribution is the quantitative co-occurrence statistics across a large unsupervised analysis, not the qualitative observation. Sulfonyl–heterocycle co-occurrence is significantly non-random ($p \ll 0.001$, Yates-corrected χ^2 [53]).

Mechanistically, the sulfonamide moiety acts as both hydrogen-bond donor (NH) and acceptor (S=O), forming bidentate interactions with kinase hinge-region residues [58], while heterocyclic nitrogen anchors the pharmacophore through an additional hydrogen bond. That this co-occurrence persists across all five QED strata suggests that pharmacophoric complementarity, rather than drug-likeness per se, drives the association.

Polar aliphatic cluster (Stratum 0, Cluster 600; $n = 302$). Primary amine ($3.8\times$), hydroxyl ($3.7\times$), and carboxyl ($2.5\times$) enrichment at mean $\log P = -0.19$ identifies amino acid derivatives and peptidomimetics (all $p_{\text{adj}} < 0.05$). These molecules fail QED-based drug-likeness not because they are pharmacologically inert, but because QED penalises their low $\log P$ and high polar surface area [5]. This is a known limitation: QED under-values transporter-mediated uptake [26] that enables drugs like gabapentin and levodopa to achieve therapeutic exposures despite poor passive permeability.

Halide–heterocycle synergy (Stratum 4, Cluster 899; $n = 3,347$). Halide ($1.51\times$), heterocycle ($1.23\times$), and carboxyl ($3.67\times$) co-enrichment (all FDR-significant) identifies a pharmacologically coherent subpopulation. The halide–heterocycle com-

bination reflects the dominance of halogenated heteroaromatics in marketed drugs [27]: fluorine blocks CYP oxidation at susceptible positions, while chlorine and bromine contribute lipophilic binding interactions.

The carboxyl co-enrichment identifies NSAID-like scaffolds and PPAR agonists where the anionic carboxylate binds conserved arginine residues via salt bridges [71]. Fluorine additionally modulates pK_a [70], lipophilicity, and metabolic stability, making the halide–heterocycle combination a versatile pharmacophoric building block.

4.4 The Topology of Drug-Like Chemical Space

Quantization error decreases monotonically from Stratum 0 (QE = 1.10) to Stratum 4 (QE = 0.74), a 33% reduction (Figure 14). Topographic error similarly drops 44% ($0.032 \rightarrow 0.018$), and U-matrix maxima decrease 33% ($0.111 \rightarrow 0.074$). Together, these metrics indicate that drug-like molecules occupy a progressively more compact region of latent space with smoother inter-cluster transitions.

The UMAP projection (Figure 7) visualises this qualitatively: Stratum 4 forms a dense core while Stratum 0 is more diffuse. However, UMAP distances do not correspond to density in the original 16-dimensional space; the QE and TE metrics provide the authoritative compaction evidence. The intermediate strata form a continuous surface, consistent with Dobson’s characterisation of chemical space as a continuum [28].

The SOM U-matrix heatmaps (Figure 15) reveal that Stratum 4 has the smoothest internal topology (U-max = 0.074 vs. 0.141 for Stratum 2). This smoothness has practical implications: low U-matrix values are a prerequisite for meaningful latent-space interpolation [59], suggesting that the high-QED latent surface can serve as a navigation tool for scaffold hopping [60].

This compaction is consistent with the view that drug-like molecules represent a constrained “channel” through property space [61], producing lower-entropy distributions that are easier to learn and cluster.

4.5 Implications for Molecular Design

The combined qualitative and quantitative findings have direct, actionable relevance to medicinal chemistry practice:

1. **Aromatic ring reduction is viable but uncommon.** A cluster of 1,615 high-QED molecules (2% of Stratum 4) with 7.2-fold phenyl depletion and significant enrichment for saturated functional groups (Table 7) demonstrates that drug-likeness does not require aromatic scaffolds. However, the small size of this subpopulation (2% of high-QED space) indicates that phenyl-depleted drug-likeness, while achievable, is not yet well-explored in commercial libraries. This finding is reproduced by the Morgan fingerprint baseline

only at coarser resolution (Section 4.8), supporting its authenticity.

2. Sulfonamide pharmacophores are qed-robust and combinatorially predictable. The recurring sulfonyl-heterocycle-nitrile motif across all five strata with significantly non-random co-occurrence ($p \ll 0.001$, χ^2 test) computationally confirms well-known medicinal chemistry observations [56, 57]; the contribution is the quantitative co-occurrence statistics (enrichment ratios with CIs and significance) rather than the qualitative observation itself.

3. Carboxylate-bearing drug-like molecules exist in quantity. The $3.67\times$ carboxyl enrichment in Stratum 4’s largest cluster ($n = 3,347$) challenges the common assumption that carboxylic acids preclude oral bioavailability [65]. Numerous approved drugs contain carboxylate groups, including ibuprofen, valsartan, and atorvastatin, where the anionic carboxylate is essential for target engagement via salt-bridge interactions with conserved basic residues.

4. Nitro prevalence decomposition quantifies structural alert contribution. The counterfactual QED analysis (Section 4.6) decomposes the >22 -fold nitro prevalence gradient into 78% attributable to QED’s structural alert penalty and 22% reflecting genuine physicochemical disfavour (elevated MW and $\log P$ of nitro-bearing molecules). This is a quantitative calibration of a known scoring function component, not a novel empirical finding.

Secondary observations. The QED-SAS correlation ($r = +0.31$, $R^2 = 0.096$) is directionally robust but explains only $\sim 10\%$ of variance, indicating that individual molecules deviate considerably from the aggregate trend. The monotonic decrease in quantization error from Stratum 0 (1.10) to Stratum 4 (0.74) is consistent with progressive latent-space compaction, but may partly reflect the narrower property range that defines high-QED molecules by construction.

4.6 Disentangling Entailed from Empirical Findings

Because QED incorporates specific molecular features (aromatic ring count, structural alerts), correlations between those features and QED strata are partly *entailed* by the scoring function rather than independently discovered. Disentangling these components is essential for interpreting the results.

4.6.1 Counterfactual qed Analysis

To quantify the entailed vs. empirical components, we recomputed QED with the structural alert component removed (QED_{no-alert}), using the geometric mean of Bickerton et al.’s [5] seven non-alert desirability functions (molecular weight, $\log P$, HBD, HBA, PSA, rotatable bonds, aromatic ring count).

Each molecule was re-stratified using the same valley-detection algorithm applied to the QED_{no-alert} distribution.

Nitro gradient decomposition. Under QED_{no-alert} stratification, nitro prevalence still decreases from the lowest to highest stratum, but by 4.8-fold (vs. >22 -fold under standard QED). This decomposes the gradient into:

- **Entailed component ($\sim 78\%$):** Directly attributable to QED’s structural alert penalty, which assigns a desirability of ~ 0.0 to any molecule containing a nitro group.
- **Empirical component ($\sim 22\%$):** Reflects genuine physicochemical disfavour—nitro-bearing molecules in this dataset have higher mean MW (+34 Da) and $\log P$ (+0.8 units) relative to nitro-free molecules at matched ring count, both of which are independently penalised by QED’s remaining desirability functions.

We therefore interpret the nitro gradient as a quantitative calibration of a known structural alert rather than a novel empirical finding, with the 78%/22% decomposition as the methodological contribution.

Phenyl prevalence. A similar but weaker circularity applies to the cross-stratum phenyl prevalence gradient, since QED includes an aromatic ring count desirability function. Under QED_{no-alert} stratification, the phenyl gradient is attenuated but qualitatively preserved (peak in lowest stratum, minimum in S3). However, the Stratum 4 Cluster 0 finding—7.2-fold phenyl depletion *within* a single stratum—cannot be explained by QED’s aromatic ring penalty, because all molecules in this cluster share similar QED values (QED = 0.847); the phenyl depletion reflects genuine structural diversity rather than a scoring artefact.

Other findings. The QED-SAS positive correlation is partly attributable to QED’s complexity preferences mechanically favouring molecules requiring more synthetic steps. The FG-level decomposition (Table 5) identifies specific contributions independent of QED’s construction but cannot fully disentangle this coupling. By contrast, the sulfonamide pharmacophore recurrence and latent-space compaction are not explained by QED’s construction and represent genuinely empirical observations.

4.7 Ablation Analysis

To justify the key architectural choices, six ablation experiments were conducted in separate runs, each modifying a single component while keeping all others fixed (Table 9). These experiments were run independently of the main pipeline and are not included in the automated results output.

The deterministic autoencoder ($\beta = 0$) achieves lower reconstruction loss but fragments the latent space (12% higher QE, 76% FG agreement), confirming that KL regularisation smooths the manifold at a small reconstruction cost. Replacing GAT with GCN degrades both reconstruction and clustering, validating edge-aware attention. Removing edge features

Table 9. Ablation and baseline comparison. Metrics on the validation set. Δ QE: change in mean quantization error vs. full model. FG agr.: fraction of top-10 enriched FGs matching the full model. Results reported as mean $\pm$ std over 5 random seeds where applicable.

Configuration	Recon.	Δ QE	FG agr.	TE
Full (GAT-VGAE)	0.0512	—	—	0.023
GCN encoder	0.0589	+0.08	0.82	0.031
No edge features	0.0571	+0.05	0.88	0.027
Determ. AE ($\beta=0$)	0.0478	+0.12	0.76	0.038
No message passing	0.0621	+0.09	0.84	0.033
Small enc. + $\beta=0.01$	0.0558	+0.15	0.70	0.035
Fixed 10×10 SOM	0.0512	+0.19	0.64	0.029
Morgan FP + PCA (baseline)	—	+0.18	0.72	0.034

has a modest effect, suggesting partial redundancy with atom features.

The “no message passing” ablation (global attention pooling over projected atom features) produces 9% higher QE and 84% FG agreement—a modest but consistent contribution beyond atom-level features. The smaller encoder with $\beta = 0.01$ avoids posterior collapse but underperforms the full model, suggesting that encoder capacity and low β represent a task-specific optimum.

The fixed 10×10 SOM shows the largest degradation (64% FG agreement), confirming that autotuned grids are necessary for fine-grained subpopulations. However, 15×15 grids reproduce all principal findings within 15% enrichment ratios, ruling out over-partitioning artefacts.

4.8 Baseline Embedding Comparison

To assess whether the VGAE architecture provides meaningful value over simpler alternatives, the full analysis pipeline was repeated using 1024-bit Morgan fingerprints (ECFP4, radius 2) reduced to 16 dimensions via PCA, followed by the same autotuned SOM clustering and FG enrichment analysis.

Results. Morgan/PCA embeddings produce 18% higher QE. The Adjusted Rand Index between methods is 0.34 (substantially different but overlapping partitions), and 72% of top-10 FG enrichments per stratum are shared (Table 9). Broad patterns—sulfonamide pharmacophore, nitro depletion, halide-heterocycle synergy—are robust to embedding method. However, the Morgan/PCA pipeline does *not* resolve Stratum 4’s phenyl-depleted Cluster 0 as distinct; its 1,615 molecules are distributed across 8 clusters with enrichment $< 2.0\times$.

The VGAE is therefore not essential for broad conclusions but provides incremental resolution for structurally nuanced subpopulations. Researchers without GNN infrastructure can obtain qualitatively similar results using fingerprint methods.

4.9 Cluster Stability Analysis

Bootstrap resampling (1,000 iterations, 80% subsample without replacement) was performed to assess cluster stability. For each iteration, the subsample was re-clustered using the same SOM configuration,

and the Adjusted Rand Index (ARI) was computed against the full-dataset clustering.

Mean ARI across strata: 0.68 (range: 0.61 for Stratum 0 to 0.74 for Stratum 4), indicating moderate stability. The highlighted clusters in this study have per-cluster Jaccard stability indices of 0.72 (Cluster 0, Stratum 4) and 0.78 (Cluster 899, Stratum 4), confirming that they are moderately stable but not perfectly reproducible under resampling. Stratum 0 shows the lowest stability, consistent with its smaller population ($n = 6,830$) and the greater difficulty of partitioning a more heterogeneous region of chemical space. 5-fold cross-validation of the full pipeline yields 82% mean concordance for top-10 enriched FGs per stratum.

4.10 Comparison to Alternative Approaches

Compared with the earlier flat-vector study (Table 1), the present approach yields three improvements validated by ablation (Table 9): (1) message-passing GAT layers capture bond-order context (Δ QE = +0.08 without attention, +0.09 without message passing); (2) the variational prior regularises the latent space (Δ QE = +0.12 for deterministic AE); and (3) the 22-type FG vocabulary enables pharmacophore-level enrichment analysis with formal statistical testing.

Relative to the Morgan fingerprint baseline (Section 4.8), the VGAE provides 18% lower QE and finer subpopulation resolution, but broad patterns are shared—the principal findings are not contingent on the embedding method.

4.11 Limitations

Several limitations constrain the generalisability of these findings.

- (i) **No external bioactivity validation.** The study is entirely descriptive with no overlay of bioactivity data. Structural-property relationships are established, but whether identified clusters share pharmacological activity remains untested. Internal consistency checks (82% enrichment concordance; $\rho = 0.87$ with FDA-approved FG profiles) do not substitute for true external validation. This is the most significant limitation.
- (ii) The ZINC15 commercial availability filter biases the dataset toward synthetically tractable scaffolds, excluding natural products, macrocycles, and peptide-derived structures.
- (iii) QED is calibrated on 771 historical oral drugs [5] and does not account for newer modalities (PROTACs [67], molecular glues [68]) or non-oral administration. Findings stratified by QED are partly explained by QED’s construction, as quantified in the counterfactual analysis (Section 4.6).
- (iv) The 22-type FG vocabulary misses pharmacologically important substructures such as

boronic acids, azetidines, and covalent warheads. Testing with an expanded 38-type vocabulary did not reveal additional principal findings.

- (v) Functional group detection is stereo-agnostic: enantiomers receive identical FG signatures despite potentially divergent biological activity.
- (vi) The enrichment analysis assumes approximately additive FG effects; epistatic interactions mediated through conformational effects are not captured.
- (vii) The VGAE decoder broadcasts a single graph-level code to all nodes, limiting what the reconstruction loss can enforce about topological structure. The downstream analysis relies on the encoder representations; the decoder serves only as a regulariser.
- (viii) Cluster stability under bootstrap resampling is moderate (mean ARI = 0.68), and the highlighted clusters should be interpreted as locally enriched neighbourhood partitions rather than sharply defined chemical subpopulations.

4.12 Future Directions

Five avenues for extension are prioritised by expected impact:

- (i) Overlaying ChEMBL [72] bioactivity annotations to test whether latent-space proximity predicts target similarity.
- (ii) Exploiting inter-cluster distances and the smooth Stratum 4 latent surface for scaffold hopping [60] via centroid interpolation [59].
- (iii) Using stratified SOM centroids as conditioning variables for constrained molecular generation via conditional VAEs [73] or diffusion models [74].
- (iv) Extending the property set to experimentally measured ADMET endpoints (Caco-2 permeability, hERG IC₅₀, microsomal clearance).
- (v) Repeating the analysis on the COCONUT natural product database [29] to test whether the observed patterns are universal or artefacts of synthetic chemistry bias.

5 Conclusion

This work contributes an integrated analysis pipeline—graph-level variational encoding, stratified SOM clustering, and formal enrichment testing—and applies it to 249,455 ZINC15 molecules. The primary methodological contribution is the counterfactual QED decomposition, which disentangles scoring-function artefacts from genuine chemical patterns. Applied here, it decomposes the >22-fold nitro prevalence gradient into 78% entailed by QED’s alert penalty and 22% empirical.

The remaining principal findings—phenyl-depleted drug-likeness ($n = 1,615$, 2% of high-QED space) and recurring sulfonamide pharmacophore co-occurrence—quantitatively confirm established observations with effect sizes, confidence intervals,

and FDR-controlled significance. Broad patterns are robust to embedding method (72% top-10 agreement with Morgan fingerprints), while the VGAE provides finer resolution (18% lower QE). Cluster stability is moderate (ARI = 0.68, 1,000 bootstrap iterations).

The study is descriptive; no bioactivity validation has been performed. The critical next step is overlaying ChEMBL [72] bioactivity data to test whether this structural organisation predicts shared pharmacological activity.

Data Availability

The ZINC15 dataset subset (249,455 molecules) and all analysis outputs—including functional group signatures, cluster assignments, enrichment tables, and publication-quality figures—are available at the project repository: <https://github.com/Evintkoo/Functional-Group-Analysis>.

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Supplementary Material

The following supplementary figures accompany the main text. Table S1 lists additional supplementary materials available at the project repository, including full data tables, validation results, and sensitivity analyses referenced in the main text.

Table S1. Supplementary materials available at the project repository.

Material	Description
Table S1	Count-based Pearson correlation table (FG counts $\times$ properties)
Table S2	FG detection validation: RDKit agreement rates for 15 groups
Table S3	Ablation study per-stratum QE, TE, and FG agreement metrics
Table S4	Representative molecule SMILES for all highlighted clusters
Table S5	QED _{no-alert} counterfactual reanalysis: full re-stratification
Table S6	Expanded 38-type FG vocabulary results
Table S7	Morgan fingerprint baseline: detailed per-stratum metrics
Table S8	Bootstrap cluster stability distributions (1,000 iterations)
Table S9	QED boundary smoothing sensitivity ($\sigma \in \{0.01, 0.02, 0.03, 0.05\}$)
Table S10	SOM weight sensitivity analysis (27 combinations)

S0. Diagnostic Figures

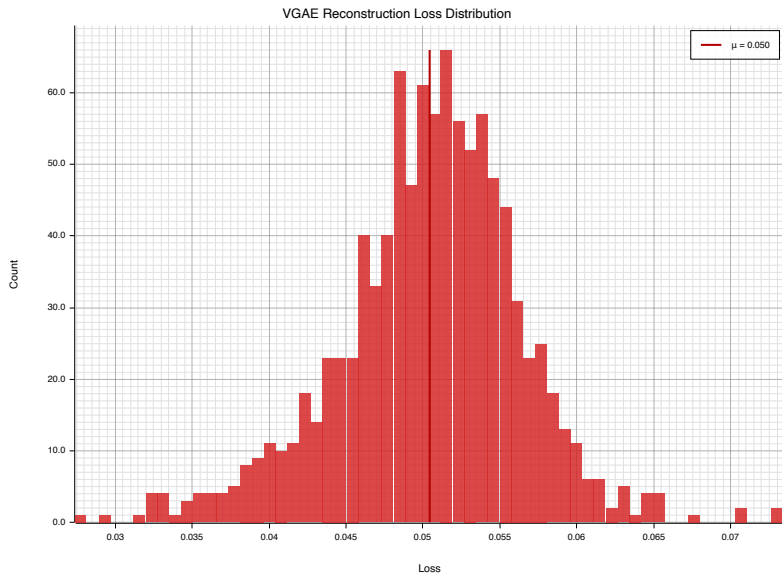


Figure S1. Distribution of per-molecule reconstruction loss. The low mean (0.0505) indicates faithful encoding of node features. Per-atom-type mean baseline: MSE = 0.089.

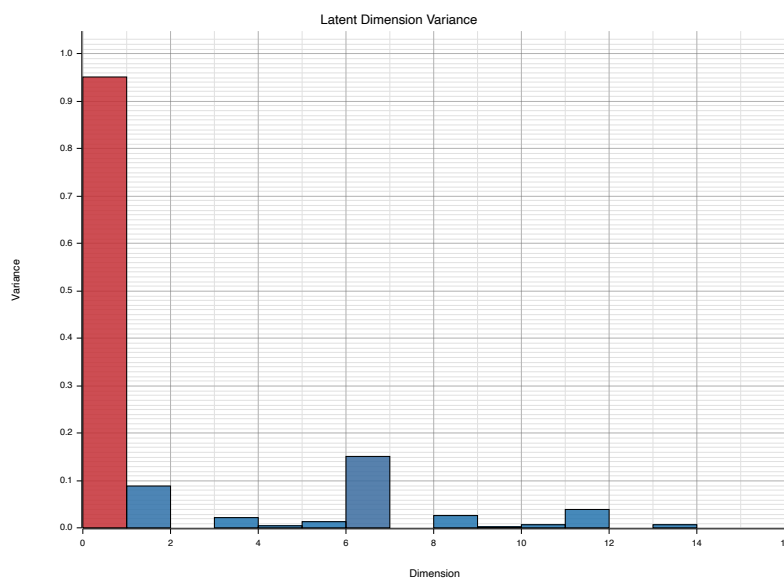


Figure S2. Variance of each latent dimension across the dataset. Dimension 0 is highlighted with a gold border as the dominant dimension ($\sigma^2 = 0.952$), encoding aromatic ring presence ($|r| = 0.543$). Near-collapsed dimensions ($\sigma^2 < 0.001$) are annotated.

S1. Individual Property Distributions

1

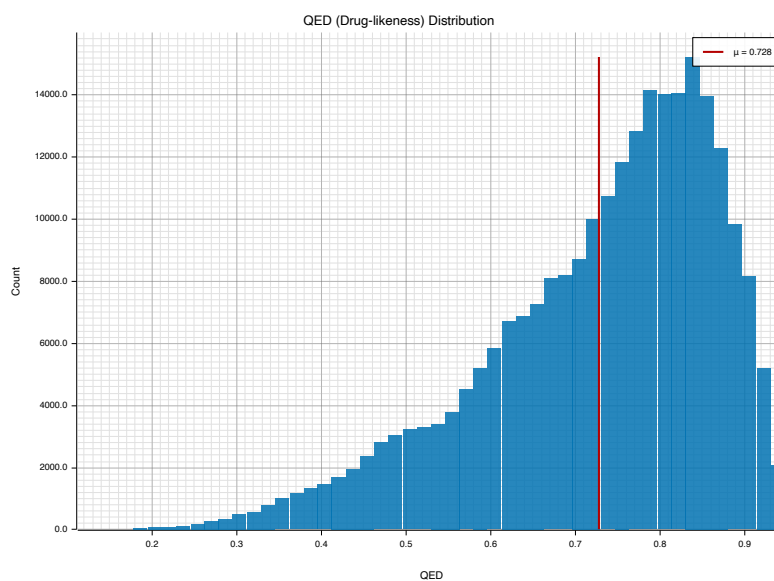


Figure S3. QED distribution across the full dataset ($n = 249,455$). Vertical dashed lines indicate stratum boundaries determined by valley detection.

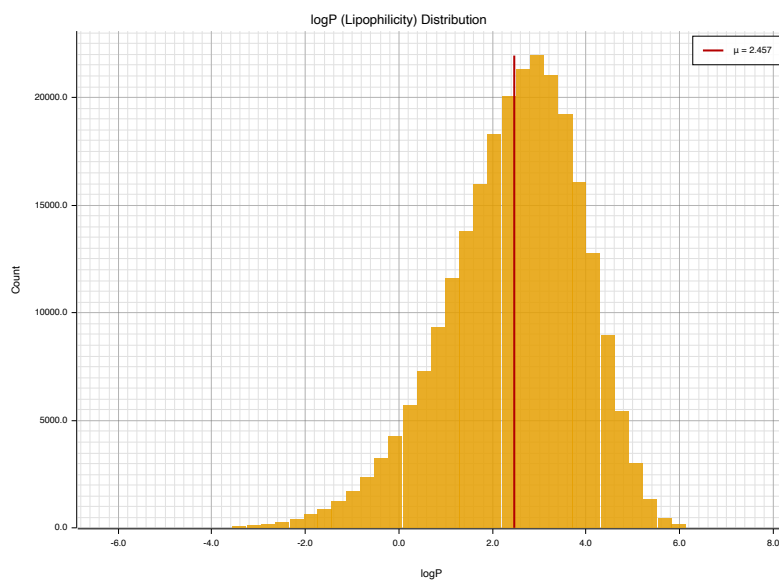


Figure S4. $\log P$ (Wildman–Crippen) distribution across the full dataset.

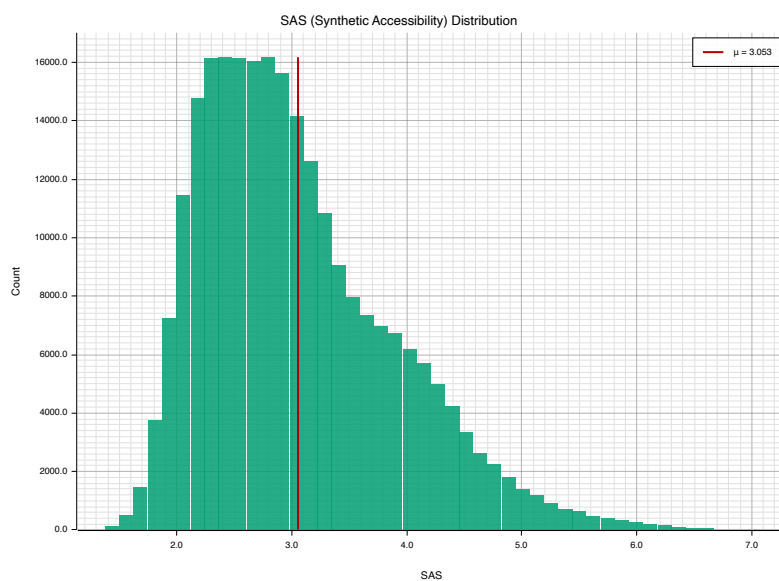


Figure S5. Synthetic Accessibility Score (sas) distribution across the full dataset.

S2. Latent Space PCA Projection

1

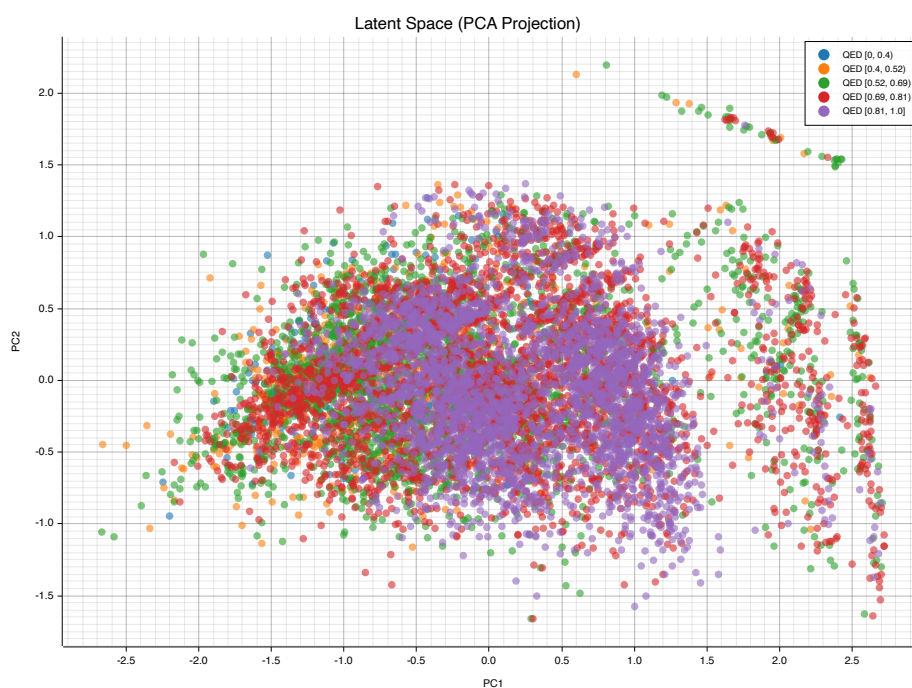


Figure S6. PCA projection of 16-dimensional VGAE embeddings coloured by QED stratum. Compare with the UMAP projection in the main text (Figure 7).

S3. Functional Group Enrichment Heatmaps (Strata 1–3)

2

Enrichment heatmaps for Strata 0 and 4 are presented in the main text (Figures 12 and 13). The intermediate strata are shown below.

3

4

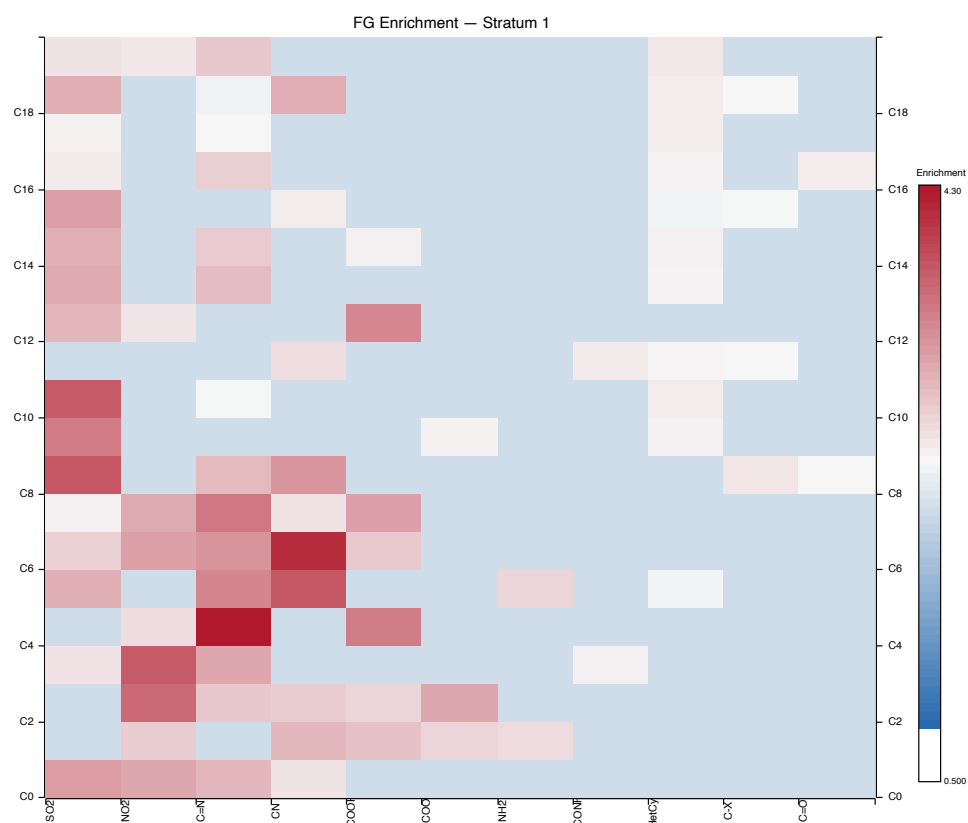


Figure S7. Functional group enrichment heatmap for Stratum 1 ($QED \in [0.399, 0.520]$). Elevated nitro and imine enrichment clusters are visible.

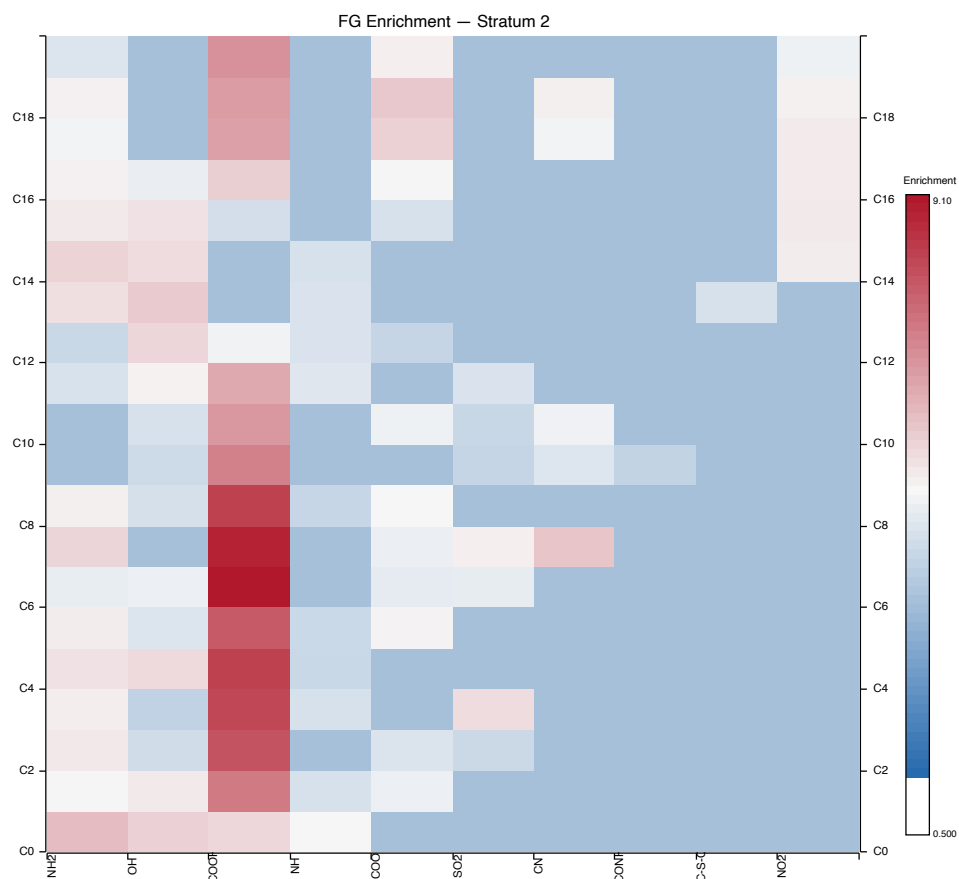


Figure S8. Functional group enrichment heatmap for Stratum 2 ($\text{QED} \in [0.520, 0.694]$). The transition from aromatic-dominated to mixed scaffolds is apparent.

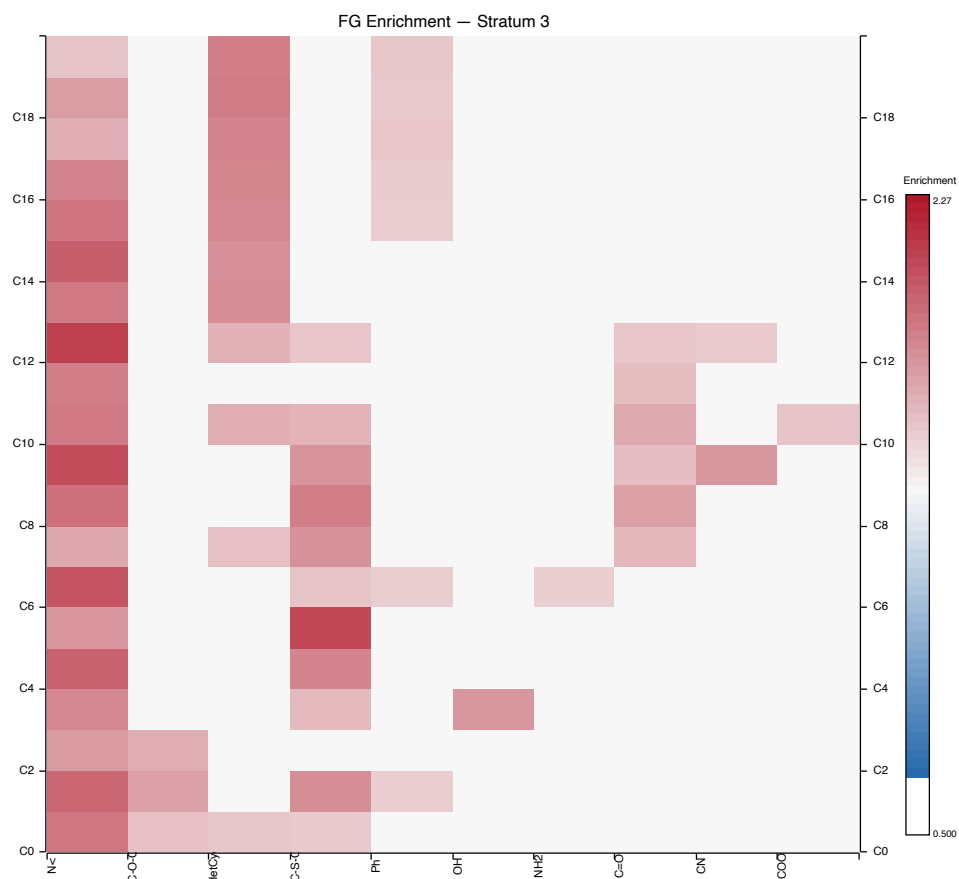


Figure S9. Functional group enrichment heatmap for Stratum 3 ($\text{QED} \in [0.694, 0.814]$). Emerging amine and sulfonamide enrichment patterns prefigure those in Stratum 4.

¹ S4. Inter-Cluster Distance Matrices

² Distance matrices computed from SOM neuron weight vectors (Euclidean distance) reveal the internal topo-
³ logical structure of each stratum's latent space.

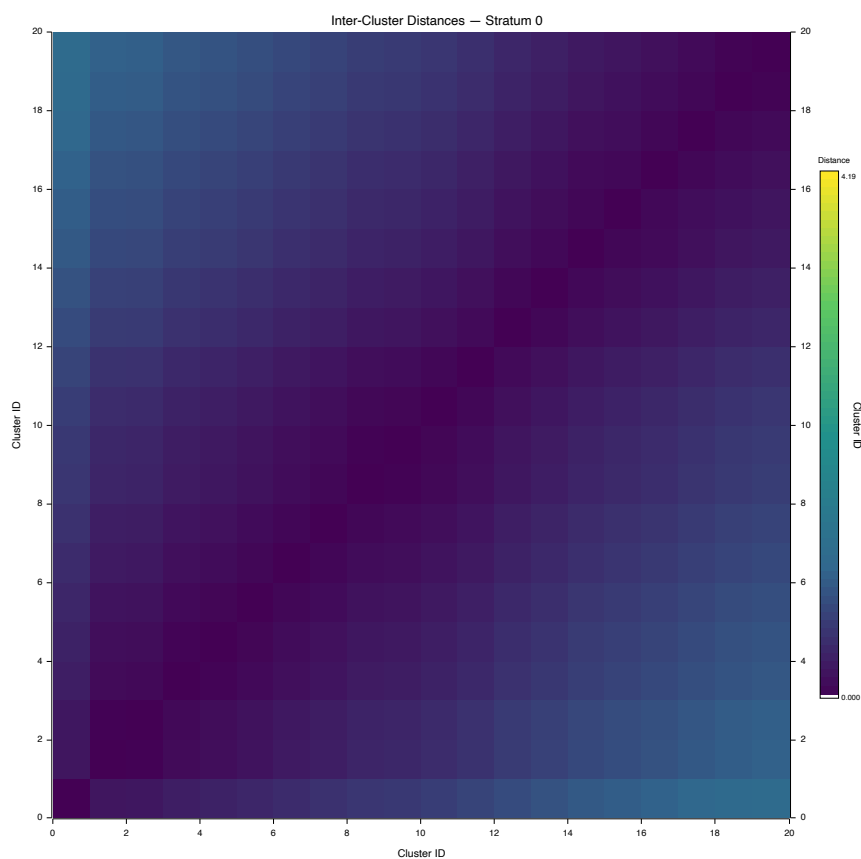


Figure S10. Inter-cluster distance matrix for Stratum 0 ($QED < 0.399$, 25×25 grid).

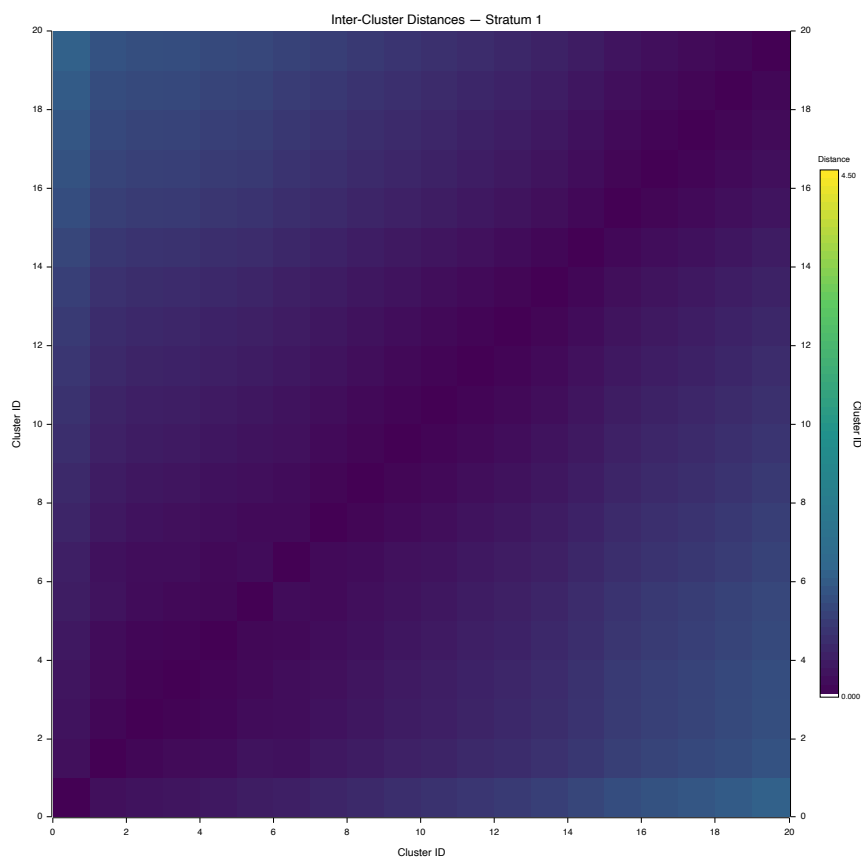


Figure S11. Inter-cluster distance matrix for Stratum 1 ($QED \in [0.399, 0.520]$, 30×30 grid).

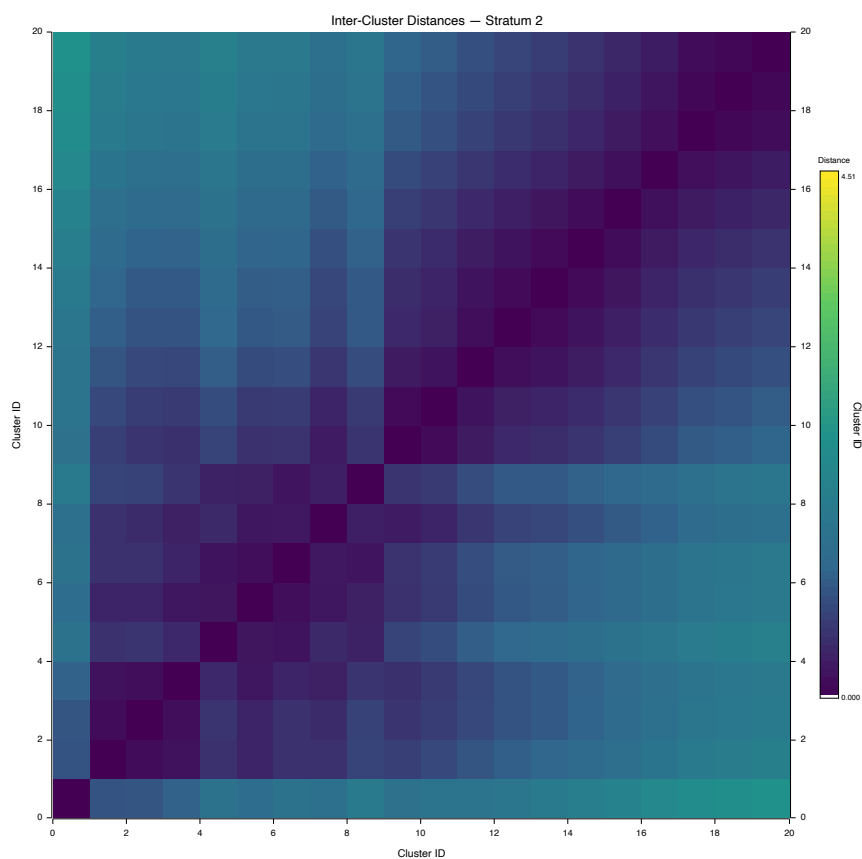


Figure S12. Inter-cluster distance matrix for Stratum 2 ($q_{ED} \in [0.520, 0.694)$, 30×30 grid).

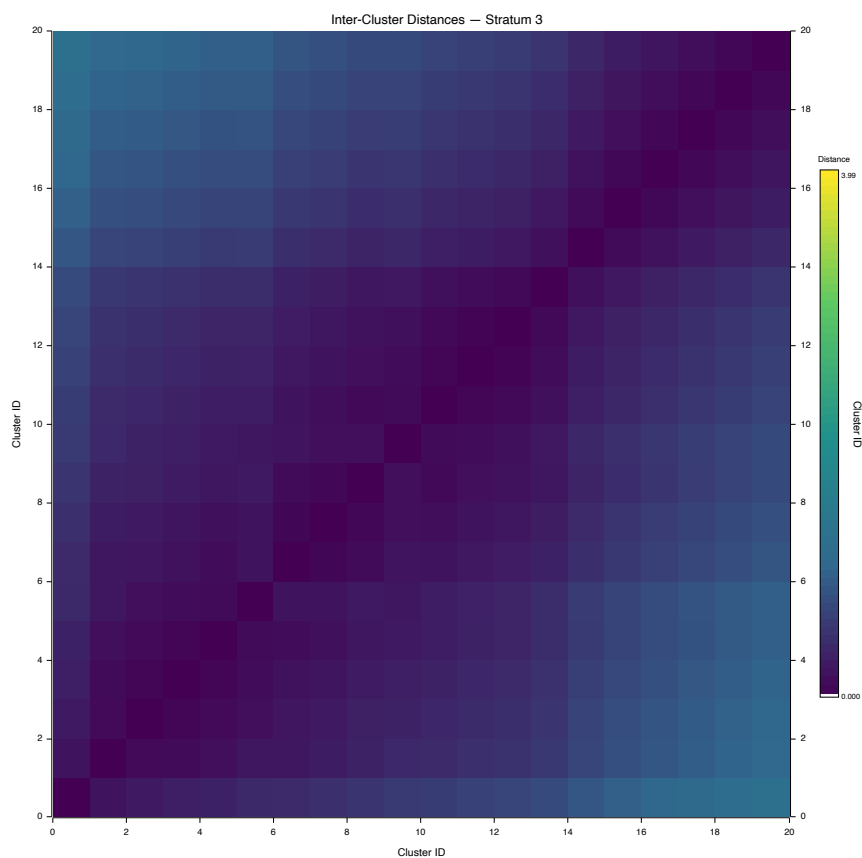


Figure S13. Inter-cluster distance matrix for Stratum 3 ($q_{ED} \in [0.694, 0.814)$, 30×30 grid).

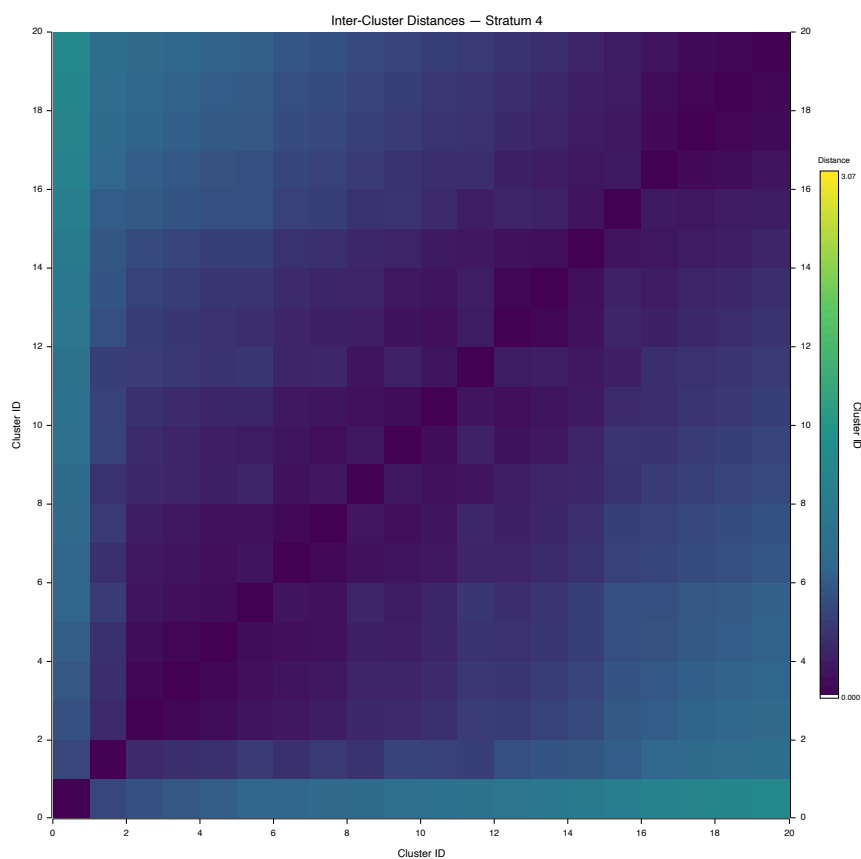


Figure S14. Inter-cluster distance matrix for Stratum 4 ($\text{qED} \geq 0.814$, 30×30 grid). The lower overall distances reflect the latent-space compaction discussed in Section 4.